SPD1657B

Advance Information

800 Source x 600 Gate
Active Matrix EPD Display Driver with Controller
for color application

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This document contains information on a new product. Specifications and information herein are subject to change without notice.

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Appendix: Revision history

Revision	Description of any change	Effective
SPD1657 Product Proposal 0.10	Initial Release for Gate 800x Source 600, comply to Eink-E2802_v06b	02-Jan-2020
SPD1657 Product Proposal 0.20	Update Command Table, Added Description	31-Mar-2020
SPD1657 Product Preview 0.10	Initial release of product preview Converted from SPD1657 Proposal 0.20 1) Updated DC and AC characteristic 2) Updated command table 3) Updated component list for application circuit 4) Updated diagram of power on/off sequence, MCU interface and IC block. 5) Removed the additional information and put into appendix	09-Sep-2020
SPD1657A Product Proposal 0.10	Initial release as the product proposal with part number as SPD1657A	08-Jan-2021
SPD1657A Product Preview 0.10	Initial release of product preview Converted from Proposal 0.10 1) Updated the figure of read/write procedure in 4 wire SPI mode 2) Updated the command description of DSP 3) Corrected the border default setting in command CDI 4) Updated Idslp_VDD (TYP) spec to 4uA 5) Remove E2802 wording in figure 13-1 6) Add requirement of slave TSDA and TSCL connection in M/S# pin description and cascade application	22-Feb-2021
SPD1657A Product Preview 0.20	1) Revise command table, remove DSP command 2) Update DC characteristic	07-Apr-2021
SPD1657A Advance Information 1.0	Convert to Advance Information Per Eink request, remove wording for 1) power scheme1 and 2, 2) ACVCOM with two scheme 3) Command table - IPC, power scheme1 and 2, VCOM sense for TFT_VCOM, GPIS command, AGID command	09-Aug-2021
SPD1657A Advance Information 1.1	1) Revised IC version as 0x03 in chip revision register 2) Updated DC characteristic. o change the condition of VSH_LV, VSH_LV2 to +15V o change the condition of VSL_LV, VSL_LV2 to -15V	24-Sep-2021
SPD1657A Advance Information 1.2	1) Added ordering part number: SPD1657BZ2 for 9um bump height 2) Updated the Typ and Max value of deep sleep current	25-Oct-2021
SPD1657A Advance Information 1.3	1) Amended revision history to include the change of Product Proposal, Product Preview and Advance Information. 2) Revised timing diagram of display on sequence from VDD power on 3) Added timing diagarm of display on sequence from deep sleep mode	27-Apr-2022
SPD1657B Product Proposal 0.10	Initial release as the product proposal with part number as SPD1657B 1) Revised timing diagram of display on sequence from VDD power on 2) Revised timing diagarm of display on sequence from deep sleep mode 3) Revised REV register	03-May-2022
SPD1657B Product Preview 0.10	Initial release as the product preview with part number as SPD1657B 1) Added Note in Display off sequence. “If VCOM and VGL set as floating after display update. Please wait until the system completely discharge before next operation.”	29-Dec-2023
SPD1657B Advance Information 1.0	Initial release as the advance Information with part number as SPD1657B	09-Jan-2024
SPD1657B Advance Information 1.1	Changed chip tray package change from size 4" x 4" to 3" x 3"	19-Sep-2024

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1 General Description

The SPD1657B is an Active Matrix EPD Display Driver with Controller which can support color application. It consists of 800 source outputs, 600 gate outputs, 2 VCOM and 1 VBD for border that can support a maximum display resolution 800x600. In addition, the SPD1657B has a cascade mode that can support higher display resolution up to display resolution 1600x600.

The SPD1657B embeds booster, regulators and oscillator. Data/Commands are sent from general MCU through the hardware selectable Serial peripheral.

2 Features

- Power supply:
  - VDD: 2.3V to 3.6V
  - VDDIO: Connect to VDD
  - VDDD/VDDDO: 1.8V, regulate from VDD supply
- Design for dot matrix type active matrix EPD display
- Resolution: 800 source outputs; 600 gate outputs; 2 VCOM; 1 VBD for border
- On chip display RAM
  - 3bit outputs per pixel to support black/ white/ color
- On-chip booster and regulator for generating VCOM, Gate and Source driving voltage
  - Gate driving output voltage:
    - VGH/VGL: +/-20V, +/-17V, +/-13V, +/-10V
    - Max 40Vp-p
  - Source / VBD driving output voltage:
    - VSH/VSL: 15V / -15V
    - VSH_LV/ VSH_LV2: 3V to 15V (Voltage step: 100mV)
    - VSL_LV/ VSL_LV2: -3V to -15V (Voltage step: 100mV)
  - VCOM output voltage
    - DCVCOM: -4V to -0.1V (Voltage step: 50mV)
    - ACVCOM: 4 levels output
      - VSH+DCVCOM, DCVCOM, VSL+DCVCOM and floating
    - Built in VCOM sensing
- 600 outputs gate driver
  - 2 levels output (VGH and VGL)
  - Up and down shift capability
- 800 outputs source driver with 3-bit resolution for black/ white/ color
  - 8 levels output (VSH, VSH_LV, VSH_LV2, GND, VSL_LV, VSL_LV2, VSL and floating)
  - Left and Right shift capability
- 1 output VBD driver for border:
  - Function same as the programmable source driver with selected LUT
  - 8 levels output (VSH+DCVCOM, VSH_LV+DCVCOM, VSH_LV2+DCVCOM, DCVCOM, VSL_LV+DCVCOM, VSL_LV2+DCVCOM, VSL+DCVCOM and floating)
- 2 output VCOM driver:
  - Support LUT
  - TFT_VCOM: DCVCOM or GND
  - FPL_VCOM: ACVCOM for 4 levels output
- On-chip oscillator
- Programmable output Waveform Settings
- Support frame rate: 100Hz (max)
- Embedded OTP to store the waveform settings and parameters
- Support low voltage detect for supply voltage
- Internal Temperature Sensor of +/-1.5degC accuracy from -25degC to 50degC, 9-bit status
- I2C single master interface to communicate with external temperature sensor
- Cascade mode to support higher display resolution
- MCU interface: Serial peripheral, Maximum SPI write speed 20MHz
  - Support 3-wire/4-wire serial peripheral
  - Support Standard 4-wire Serial SPI, Dual SPI and Quad SPI.
- Low current consumption for operation and sleep
- Available in COG package

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3 ORDERING INFORMATION

Table 3-1 : Ordering Information

Ordering Part Number	Package Form	Remark
SPD1657BZ2	Gold Bump Die	Bump Face Up On Waffle pack Die thickness: 300um Bump height: 9um

4 Block Diagram

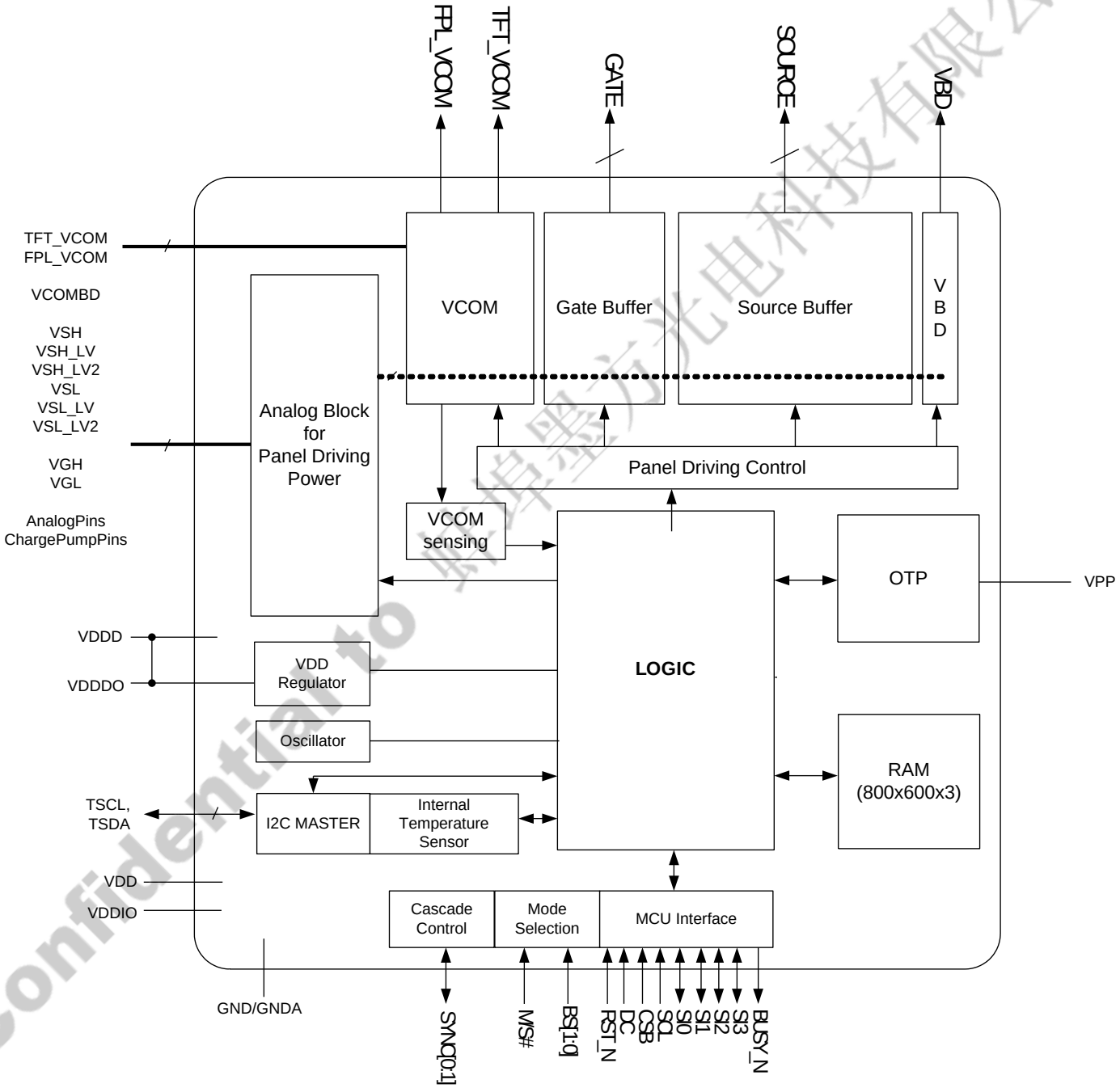


Figure 4-1 : SPD1657B Block Diagram

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5 PIN DESCRIPTION

Key: I = Input, O =Output, IO = Bi-directional (input/output), P = Power pin, C = Capacitor Pin
NC = Not Connected, Pull L =connect to GND, Pull H = connect to V_{DDIO}

Table 5-1: Power Supply Pins

Name	Type	Connect to	Function	Description	When not in use
VDD	P	Power Supply	Power Supply	Power input pin for the chip.	-
VDDIO	P	Power Supply	Power for interface logic pins	Power input pin for the interface. - Connect to VDD in the application circuit.	-
VDDDO	C	Capacitor	Regulator output	Core logic power regulator output pin. VDDDO is regulated internally from VDD.	-
VDDD	P	Power Supply	Power for logic pins	Core logic power pin - connect to VDDDO in the application circuit	-
GND	P	GND	GND	Ground	-
GNDA	P	GND	GND	Ground - Connect to GND in the application circuit	-
VPP	P	Reserved	Reserved	Reserved Keep it floating.	-

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Table 5-2: Interface Logic Pins

Name	Type	Connect to	Function	Description	When not in use
SCL	I	MPU	Data Bus	Serial clock pin for interface	-
SI[0:3]	I/O	MPU	Data Bus	Serial data pin for interface	VDDIO or GND
CSB	I	MPU	Logic Control	This pin is the chip select input connecting to the MCU.	VDDIO or GND
DC	I	MPU	Logic Control	This pin is Data/Command control pin connecting to the MCU.	VDDIO or GND
RST_N	I	MPU	System Reset	This pin is reset signal input. Active Low.	-
BUSY_N	O	MPU	Device Busy Signal	This pin indicates the driver status. L: Driver is busy H: non-busy. Host side can send command/data to driver. In the cascade mode, the BUSY pin of the slave chip should be left open.	Open
M/S#	I	VDDIO/GND	Cascade Mode Selection	This pin is Master and Slave selection pin. - For the single chip application, the M/S# pin should be connected to VDDIO. - In the cascade mode: For Master Chip, the M/S# pin should be connected to VDDIO. For Slave Chip, the M/S# pin should be connected to GND. The oscillator, booster and regulator circuits of the slave chip will be disabled. The corresponding pins including TSDA, TSCL, SYNC[0:1], VDDD, VGH, VGL, VSH, VSL, VSH_LV, VSL_LV, VSH_LV2, VSL_LV2, TFT_VCOM, FPL_VCOM and VCOMBD must be connected to the master chip.	-
BS[1:0]	I	VDDIO/GND	MCU Interface Mode Selection	This pin is for selecting 3-wire or 4-wire SPI bus. BS[1:0] = 00: 4-wire SPI BS[1:0] = 01: 3-wire SPI(9 bits SPI) BS[1:0] = 10 or 11: Standard 4-wire SPI/Dual SPI/Quad SPI	-
TSDA	I/O	Temperature sensor SDA	Interface to Digital Temp. Sensor	This pin is I^2C Interface to digital temperature sensor Data pin. External pull up resistor is required when connecting to I^2C slave.	GND
TSCL	O	Temperature sensor SCL	Interface to Digital Temp. Sensor	This pin is I^2C Interface to digital temperature sensor Clock pin. External pull up resistor is required when connecting to I^2C slave.	GND

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Table 5-3: Display Power Pins

Name	Type	Connect to	Function	Description	When not in use
VGH	C	Stabilizing capacitor	Gate High Power	This pin is Positive Gate driving voltage.	-
VGL	C	Stabilizing capacitor	Gate Low Power	This pin is Negative Gate driving voltage.	-
VSH	C	Stabilizing capacitor	Source power	This pin is Positive Source driving voltage, VSH Connect a stabilizing capacitor between VSH and GND in the application circuit.	-
VSH_LV	C	Stabilizing capacitor	Source power	This pin is Positive Source driving voltage, VSH_LV Connect a stabilizing capacitor between VSH_LV and GND in the application circuit.	-
VSH_LV2	C	Stabilizing capacitor	Source power	This pin is Positive Source driving voltage, VSH_LV2 Connect a stabilizing capacitor between VSH_LV2 and GND in the application circuit.	-
VSL	C	Stabilizing capacitor	Source power	This pin is Negative Source driving voltage, VSL. Connect a stabilizing capacitor between VSL and GND in the application circuit.	-
VSL_LV	C	Stabilizing capacitor	Source power	This pin is Negative Source driving voltage, VSL_LV. Connect a stabilizing capacitor between VSL_LV and GND in the application circuit.	-
VSL_LV2	C	Stabilizing capacitor	Source power	This pin is Negative Source driving voltage, VSL_LV2. Connect a stabilizing capacitor between VSL_LV2 and GND in the application circuit.	-
VCOMBD	C	Stabilizing capacitor	VBD power	This pin is VCOMBD driving voltage. Connect a stabilizing capacitor between VCOMBD and GND in the application circuit.	-

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Table 5-4: Analog Pins

Name	Type	Connect to	Function	Description	When not in use
CP1N CP1P CP2N CP2P VCP1 VCP2	C	Charge Pump Pin	Reserved	Reserved	-
GDRC GDRH RESEC RESEH	I/O	Analog Pin	Reserved	Reserved	-
VPC	C	Stabilizing capacitor	Reserved	Reserved	-
REFN REFP	C	Reserved	Reserved	Reserved	-

Table 5-5: Driver Output Pins

Name	Type	Connect to	Function	Description	When not in use
S [799:0]	O	Panel	Source driving signal	Source output pin.	Open
G [599:0]	O	Panel	Gate driving signal	Gate output pin.	Open
VBD	O	Panel	Border driving signal	Border output pin.	Open
FPL_VCOM	C	Panel & Stabilizing capacitor	VCOM driving signal	VCOM output pin for FPL Connect a stabilizing capacitor between FPL_VCOM and GND in the application circuit.	-
TFT_VCOM	C	Panel & Stabilizing capacitor	VCOM driving signal	VCOM output pin for TFT Connect a stabilizing capacitor between TFT_VCOM and GND in the application circuit.	-

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Table 5-6: Other Pins

Name	Type	Connect to	Function	Description	When not in use
SYNC[0:1]	I/O	Open	Reserved	Reserved for cascade application.	Open
GPIO [0:1]	I/O	Open	Reserved	Reserved	Open
TIN [1:2]	I	GND	Reserved	Reserved	GND
TPA [1:5] TNA [1:3]	O	Open	Reserved	Reserved	Open
TPL [1:4]	O	Open	Reserved	Reserved	Open
CGH1N CGH1P	C	Open	Reserved	Follow application circuit.	-
GDRL RESEL	I/O	Analog Pin	Reserved	Follow application circuit.	-
VPH	C	Analog Pin	Reserved	Follow application circuit.	-

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6 Functional Block Description

6.1 MCU Interface

6.1.1 MCU Interface selection

The SPD1657B can support 3-wire/4-wire serial peripheral. In the SPD1657B, the MCU interface is pin selectable by BS[1:0] shown in Table 6-1.

In standard 4-wire serial peripheral, it is command selectable for Serial SPI, Dual SPI and Quad SPI interface.

Note

(1) L is connected to GND

(2) H is connected to V_{DDIO}

Table 6-1 : Interface pins assignment under different MCU interface

	Pin Name						
MCU Interface	**BS1**	**BS0**	**RST_N**	**CSB**	**DC**	**SCL**	**SI[0:3]**
4-wire serial peripheral interface (SPI)	L	L	Required	CSB	D/C	SCL	SI0=SDA
3-wire serial peripheral interface (SPI) – 9 bits SPI	L	H	Required	CSB	L	SCL	SI0=SDA
Standard 4-wire SPI	H	L/H	Required	CSB	L	SCL	

	Serial SPI	Dual SPI	Quad SPI
SI0	MOSI	MOSI	MOSI
SI1	MISO	SI1	SI1
SI2	L	L	SI2
SI3	L	L	SI3

6.1.2 MCU Serial Peripheral Interface (4-wire SPI)

The 4-wire SPI consists of serial clock SCL, serial data SDA, D/C and CSB. The control pins status in 4-wire SPI in reading/writing command/data is shown in

Table 6-2.

The read/write procedure of 4-wire SPI is shown in Figure 6-1.

Table 6-2 : Control pins status of 4-wire SPI

Function	SCL pin	SDA pin	D/C pin	CSB pin
Write command	↑	Command bit	L	L
Read/Write data	↑	Data bit	H	L

Note:

(1) L is connected to GND and H is connected to V_{DDIO}

(2) ↑ stands for rising edge of signal

(3) SDA (Write Mode) is shifted into an 8-bit shift register on every rising edge of SCL in the order of D7, D6, ... D0. The level of D/C should be kept over the whole byte. The data byte in the shift register is written to the Graphic Display Data RAM (RAM)/Data Byte register or command Byte register according to D/C pin.

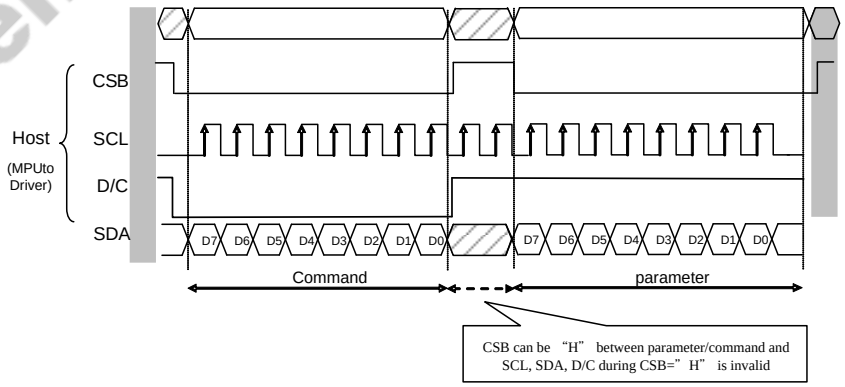


Figure 6-1 : Read/Write procedure in 4-wire SPI mode

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6.1.3 MCU Serial Peripheral Interface (3-wire SPI)

The 3-wire SPI consists of serial clock SCL, serial data SDA and CSB. The operation is similar to 4-wire SPI while D/C pin is not used and it must be tied to LOW. The control pins status in 3-wire SPI is shown in Table 6-3.

In the read/write operation, a 9-bit data will be shifted into the shift register on every clock rising edge. The bit shifting sequence is D/C bit, D7 bit, D6 bit to D0 bit. The first bit is D/C bit which determines the following byte is command or data. When D/C bit is 0, the following byte is command. When D/C bit is 1, the following byte is data. The read/write procedure of 3-wire SPI is shown in Figure 6-2.

Table 6-3 : Control pins status of 3-wire SPI

Function	SCL pin	SDA pin	D/C pin	CSB pin
Write command	↑	Command bit	Tie LOW	L
Read/Write data	↑	Data bit	Tie LOW	L

Note:

(1) L is connected to GND and H is connected to V_{DDIO}

(2) ↑ stands for rising edge of signal

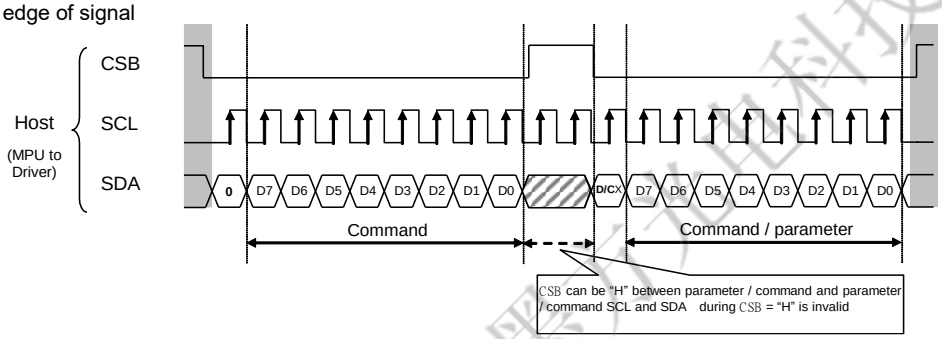


Figure 6-2 : Read/Write procedure in 3-wire SPI mode

6.1.4 Standard 4-wire SPI, for Serial SPI, Dual SPI and Quad SPI

The Standard 4-wire SPI consists of serial clock SCL, serial data input SI0 (MISO), output SI1(MOSI) and CSB.
The Dual SPI consists of serial clock SCL, serial data SI0, SI1 and CSB.
The Quad SPI consists of serial clock SCL, serial data SI0, SI1, SI2, SI3 and CSB.
When Dual / Quad SPI is selected, SI0, SI1 has two roles. Multiple Data link is ONLY for RAM access command.

After CSB pull low, command byte of an 8-bit data will be shifted into the shift register on every clock rising edge. The bit shifting sequence is D7 bit, D6 bit to D0 bit. Then following byte is data. The data format shows the procedure of Serial SPI in Figure 6-3, Dual SPI in Figure 6-4, the Quad SPI (Write Mode) in Figure 6-5, and the Quad SPI (Read Mode) in Figure 6-6.

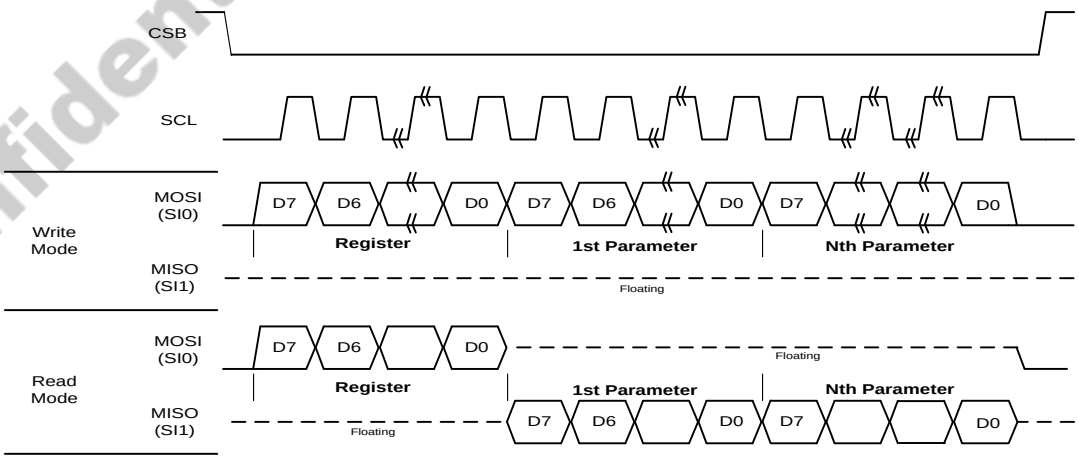


Figure 6-3 : Operation in Serial SPI

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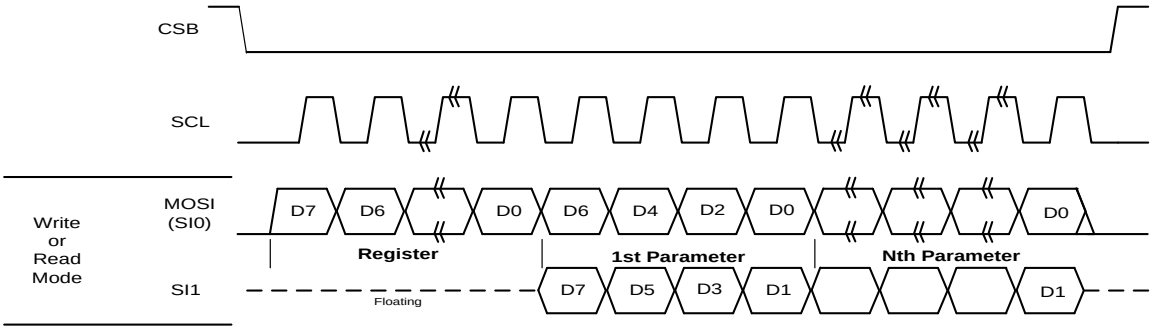


Figure 6-4 : Operation in Dual SPI

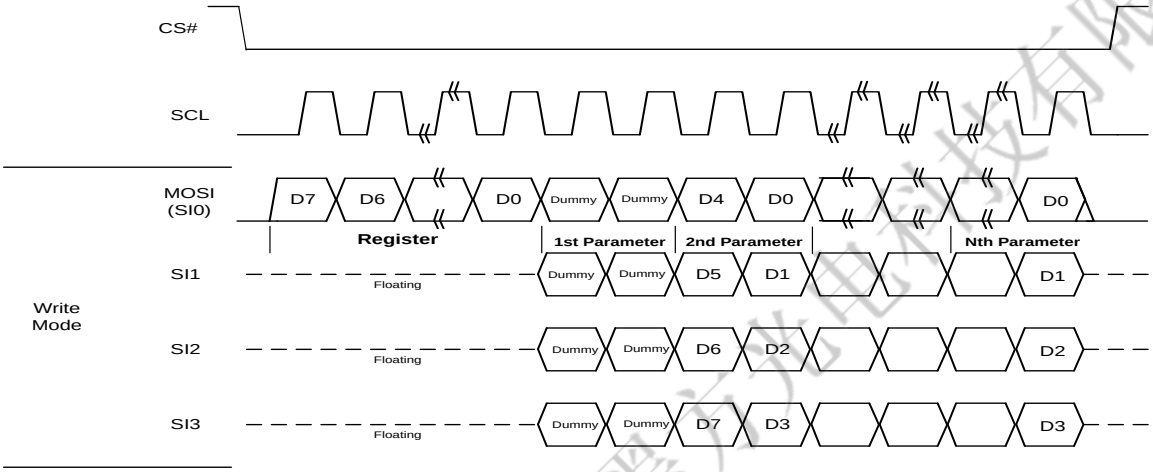


Figure 6-5 : Operation in Quad SPI (Write Mode)

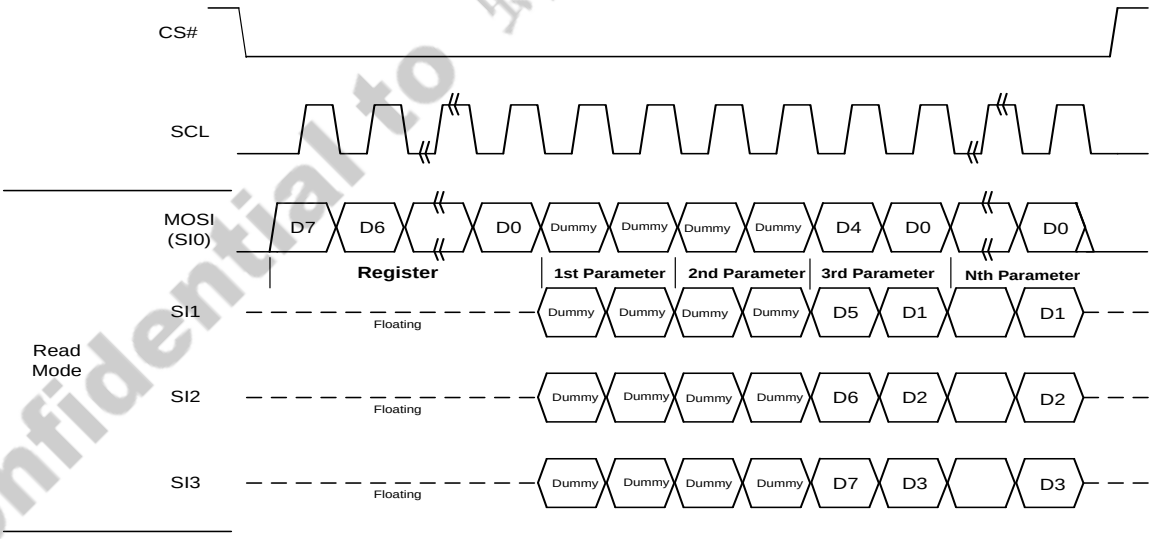


Figure 6-6 : Operation in Quad SPI (Read Mode)

6.2 RAM

The On chip display RAM is holding the image data. 3bpp for 800x600 resolution

6.3 Oscillator

The oscillator module generates the clock reference for waveform timing and analog operations.

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6.4 Booster & Regulator

A voltage generation system is included in the driver. It provides all necessary driving voltages required for an AMEPD panel including VGH, VGL, VSH, VSL, VSH_LV, VSL_LV, VSH_LV2, VSL_LV2, TFT_VCOM, FPL_VCOM and VCOMBD. External application circuit is needed to make the on-chip booster & regulator circuit work properly.

6.5 VCOM Sensing

This functional block provides the scheme to select the optimal VCOM DC level. The sensed value can be programmed into OTP. It required PON with DCVCOM mode for analog voltage ready.
The typical flow of VCOM sensing:

- Active Gate is scanning during the VCOM sense Period.
- Source are GND.
- FPL_VCOM pin is used for sensing.
- During Sensing period, BUSY_N is kept flagged.
- After Sensing, Active Gate return to non-select stage.

6.6 External Temperature Sensor I2C Single Master Interface

The chip provides two I/O lines [TSDA and TSCL] for connecting digital temperature sensor for temperature reading sensing.
TSDA will treat as SDA line and TSCL will treat as SCL line. They are required connecting with external pull-up resistor.

1. If the Temperature value MSByte bit D11 = 0, then
   the temperature is positive and value (DegC) = + (Temperature value) / 16
2. If the Temperature value MSByte bit D11 = 1, then
   the temperature is negative and value (DegC) = - (2's complement of Temperature value) / 16

12-bit binary (2's complement)	Hexadecimal Value	Decimal Value	Value [DegC]
1100 1001 0000	C90	-880	-55
1100 1001 0010	C92	-878	-54.875
1110 0111 0000	E70	-400	-25
1111 1111 1110	FFE	-2	-0.125
0000 0000 0000	0	0	0
0000 0000 0010	2	2	0.125
0001 1001 0000	190	400	25
0111 1101 0000	7D0	2000	125
0111 1110 0010	7E2	2018	126.125
0111 1110 1110	7EE	2030	126.875
0111 1111 0000	7F0	2032	127

6.7 Cascade Mode

The SPD1657B has a cascade mode that can cascade 2 chips to achieve the display resolution up to 1600 (sources) x 600 (gates). The pin M/S# is used to configure the chip. When M/S# is connected to VDDIO, the chip is configured as a master chip. When M/S# is connected to GND, the chip is configured as a slave chip.

When the chip is configured as a master chip, it will be the same as a single chip application, ie, all circuit blocks will be worked as usual. When the chip is configured as a slave chip, its oscillator, booster and regulator circuit will be disabled. The oscillator clock and all booster voltages will come from the master chip. Therefore, the corresponding pins including TSDA, TSCL, SYNC[0:1], VDDD, VGH, VGL, VSH, VSL, VSH_LV, VSL_LV, VSH_LV2, VSL_LV2, TFT_VCOM, FPL_VCOM and VCOMBD must be connected to the master chip.

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7 COMMAND TABLE

7.1 User Command Table

W/R: 0: Write Cycle 1: Read Cycle C/D: 0: Command / 1: Data D7~D0: –: Don't Care #: Valid Data

Command	R/W	D/CX	D7	D6	D5	D4	D3	D2	D1	D0	Registers	
PSR	W	0	0	0	0	0	0	0	0	0		00H
	W	1	#	#	0	#	#	#	#	#	RES[1:0], 0, P_mode, UD, SHL, SHD_N, RST_N	
	W	1	#	#	#	#	#	#	#	#	LUT_EN, FOPT[1:0], VCMZ, TS_AUTO, TIEG, NORG, VC_LUTZ	
PWR	W	0	0	0	0	0	0	0	0	1		01H
	W	1	-	-	1	1	#	#	#	#	VSC2_EN, VSC_EN, VS_EN, VG_EN	
	W	1	-	-	-	-	-	-	#	#	VGPN[1:0]	
	W	1	-	#	#	#	#	#	#	#	VSPL[6:0]	
	W	1	-	#	#	#	#	#	#	#	VSNL[6:0]	
	W	1	-	#	#	#	#	#	#	#	VSPL2[6:0]	
	W	1	-	#	#	#	#	#	#	#	VSNL2[6:0]	
POF	W	0	0	0	0	0	0	0	1	0		02H
	W	1	-	-	-	-	-	-	-	0		
POFS	W	0	0	0	0	0	0	0	1	1		03H
	W	1	0	0	#	#	#	#	#	#	T_VDPG_OFF[1:0], T_VDS_ON[1:0], T_VDS_OFF[1:0]	
	W	1	0	1	0	1	0	1	0	0		
	W	1	0	0	0	0	0	0	0	0		
	W	1	0	1	0	0	0	1	0	0		
PON	W	0	0	0	0	0	0	1	0	0		04H
BTST1	W	0	0	0	0	0	0	1	0	1		05H
	W	1	0	#	#	#	#	#	#	#	VPCSSC_ONT[2:0], T_VPCSSA[1:0] , T_VPCSSB[1:0]	
	W	1	0	#	#	#	#	#	#	#	VPC_SSA_DRV[2:0], VPCSSA_OFFT[3:0]	
	W	1	0	#	#	#	#	#	#	#	VPC_SSB_DRV[2:0], VPCSSB_OFFT[3:0]	
	W	1	0	#	#	#	#	#	#	#	VPC_SSC_DRV[2:0], VPCSSC_OFFT[3:0]	
BTST2	W	0	0	0	0	0	0	1	1	0		06H
	W	1	0	#	#	#	#	#	#	#	VPHSSC_ONT[2:0], T_VPHSSA[1:0] , T_VPHSSB[1:0]	
	W	1	0	#	#	#	#	#	#	#	VPH_SSA_DRV[2:0], VPHSSA_OFFT[3:0]	
	W	1	0	#	#	#	#	#	#	#	VPH_SSB_DRV[2:0], VPHSSB_OFFT[3:0]	
	W	1	0	#	#	#	#	#	#	#	VPH_SSC_DRV[2:0], VPHSSC_OFFT[3:0]	
DSLP	W	0	0	0	0	0	0	1	1	1		07H
	W	1	1	0	1	0	0	1	0	1	Check code	
DTM	W	0	0	0	0	1	0	0	0	0		10H
	W	1	-	#	#	#	-	#	#	#	KPixel1[2:0], KPixel2[2:0]	
	W	1	..	..	..	..	..	..	..	..	…	
	W	1	-	#	#	#	-	#	#	#	Kpixel[2M-1][2:0], Kpixel[2M][2:0]	
	W	1	#	#	#	#	#	#	#	#	Kpixel1[1:0], KPixel2[1:0], KPixel3[1:0], KPixel4[1:0]	
	W	1	..	..	..	..	..	..	..	..	…	
	W	1	#	#	#	#	#	#	#	#	Kpixel[2M-3][1:0], Kpixel[2M-2][1:0], Kpixel[2M-1][1:0], Kpixel[2M][1:0]	
DRF	W	0	0	0	0	1	0	0	1	0		12H
	W	1	-	-	-	-	-	-	-	#	AC/DCVCOM	
AUTO	W	0	0	0	0	1	0	1	1	1		17H
	W	1	#	#	#	#	#	#	#	#	Check code	
PLL	W	0	0	0	1	1	0	0	0	0		30H
	W	1	-	-	-	-	0	#	#	#	FR[2:0]	
TSC	W	0	0	1	0	0	0	0	0	0		40H
	R	1	#	#	#	#	#	#	#	#	D[10:3] / TS[7:1]	
	R	1	#	#	#	-	-	-	-	-	D[2:0] / TS[0]	
TSE	W	0	0	1	0	0	0	0	0	1		41H
	W	1	#	-	-	-	#	#	#	#	TSE, TO[3:0]	
TSW	W	0	0	1	0	0	0	0	1	0		42H
	W	1	#	#	#	#	#	#	#	#	WATTR[7:0]	
	W	1	#	#	#	#	#	#	#	#	WMSB[7:0]	
	W	1	#	#	#	#	#	#	#	#	WLSB[7:0]	
TSR	W	0	0	1	0	0	0	0	1	1		43H
	R	1	#	#	#	#	#	#	#	#	RMSB[7:0]	
	R	1	#	#	#	#	#	#	#	#	RLSB[7:0]	
CDI	W	0	0	1	0	1	0	0	0	0		50H
	W	1	#	#	#	#	#	#	#	#	VBD[2:0], DDX, CDI[3:0]	
LPD	W	0	0	1	0	1	0	0	0	1		51H
	R	1	-	-	-	-	-	-	-	#	LPD	
TRES	W	0	0	1	1	0	0	0	0	1		61H
	W	1	-	-	-	-	-	-	#	#	HRES[9:2]	
	W	1	#	#	#	#	#	#	0	0		
	W	1	-	-	-	-	-	-	#	#	VRES[9:0]	
	W	1	#	#	#	#	#	#	#	#		
REV	W	0	0	1	1	1	0	0	0	0		70H
	R	1	0	0	0	0	0	0	0	1	01H	
	R	1	#	#	#	#	#	#	#	#	REV1[7:0]	
	R	1	#	#	#	#	#	#	#	#	REV2[7:0]	
AMV	W	0	1	0	0	0	0	0	0	0		80H
	W	1	#	#	#	#	#	#	#	#	P[1:0], AMVT[1:0], AMVX, AMVS, AMV,AMVE	
	W	1	-	-	-	-	-	-	1	0	02H	

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VV	W	0	1	0	0	0	0	0	0	1		81H
	R	1	-	#	#	#	#	#	#	#	VV[6:0]	
VDCS	W	0	1	0	0	0	0	0	1	0		82H
	W	1	-	#	#	#	#	#	#	#	VDCS[6:0]	
PTLW	W	0	1	0	0	0	0	0	1	1		83H
	W	1	-	-	-	-	-	-	#	#	HRST[9:2]	
	W	1	#	#	#	#	#	#	0	0		
	W	1	-	-	-	-	-	-	#	#	HRED[9:2]	
	W	1	#	#	#	#	#	#	0	0		
	W	1	-	-	-	-	-	-	#	#	VRST[9:0]	
	W	1	#	#	#	#	#	#	#	#		
	W	1	-	-	-	-	-	-	#	#	VRED[9:0]	
	W	1	#	#	#	#	#	#	#	#		
	W	1	-	-	-	-	-	-	-	#	PMODE	
T_VDCS	W	0	1	0	0	0	0	1	0	0		84H
	W	1	-	-	-	-	-	-	-	#	T_VC_S	
PGM	W	0	1	0	0	1	0	0	0	0		90H
APG	W	0	1	0	0	1	0	0	0	1		91H
ROTP	W	0	1	0	0	1	0	0	1	0		92H
	R	1	#	#	#	#	#	#	#	#	Dummy	
	R	1	#	#	#	#	#	#	#	#	OTP 1st byte	
	R	1	#	#	#	#	#	#	#	#	…	
	R	1	#	#	#	#	#	#	#	#	OTP Nth byte	
PWS	W	0	1	1	1	0	0	0	1	1		E3H
	W	1	#	#	#	#	#	#	#	#	VCOM_W[3:0], SD_W[3:0]	
LVSEL	W	0	1	1	1	0	0	1	0	0		E4H
	W	1	-	-	-	-	-	-	#	#	LVD_SEL[1:0]	
SPIM	W	0	1	1	1	0	0	1	0	1		E5H
	W	1	-	-	-	-	-	-	#	#	SM[1:0]	

Note:

(1) Any bits shown here as 0 must be written with a 0. All unused bits should also be set to zero. Device malfunction may occur if this is not done.

(2) Commands are effective ONLY when ALL parameters corresponding to that command is written.

(3) Registers marked 'W/R' can be read, but the contents are written when the SPI command completes – so the contents can be read and altered. The user can subsequently write the register to restore the contents following an SPI read.

(4) All write commands are "UNAVAILABLE" when BUSY_N=0 is asserted by PSR (R00h) or DSLP (R07h) or DRF (R12h) or AUTO (R17h) or PON (R04h) or POF (R02h) or TSC (R40h) or TSW (R42h) or TSR (R43h) or LPD (R51h) or TRES (R61h) or AMV (R80h) or PGM (R90h) or APG (R91h). All read commands are always "AVAILABLE".

* AVAILABLE means that Host can send command/parameter to driver.

* UNAVAILABLE means that Host cannot send command/parameter to driver.

(5) Commands or Parameter not shown in above table are reserved.

(6) All write and read commands are only valid after normal power on sequence

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8 COMMAND DESCRIPTION

R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description
0	0	00	0	0	0	0	0	0	0	0	PSR	Panel Setting Register A[7:0] = 0Fh [POR] B[7:0] = 09h [POR]
0	1		A_7	A_6	0	A_4	A_3	A_2	A_1	A_0		A[7:6] ~ RES[1:0] Display Resolution setting (source x gate) 00b: 800 x 600 (Default) 01b: 800 x 480 10b: 600 x 480 11b: 600 x 448 *Remark: Inactive Gate outputs will be VGL for DRF and AMV. Inactive Source outputs will follow VCOM LUT output for DRF A[5] = 0, If VCOMH>= VGH-2V, VCOMH setting will be remapped as VGH-2V If VCOML<= VGL+2V, VCOML setting will be remapped as VGL+2V A[4] ~ P_mode. Pixel Mode 0: 2-bit/pixel (Default) 1: 3-bit/pixel A[3] ~ UD Gate Scan Direction: 0: Scan down. First line to Last line: Gn-1 … G0 1: Scan up. (Default) First line to Last line: G0 … Gn-1 A[2] ~ SHL Source Shift Direction: 0: Shift left. First data to Last data: Sn-1 … S0 1: Shift right. (Default) First data to Last data: S0 … Sn-1 A[1] ~ SHD_N Booster and Regulator Switch: 0: PON / POF command will not execute 1: PON / POF command will execute (Default) A[0] ~ RST_N Soft Reset: 0: The controller is reset. Reset all registers to their default value. Driver all function will be disabled. 1: Normal operation (Default). BUSY_N signal will become “0” until Soft reset is finished.
0	1		B_7	B_6	B_5	B_4	B_3	B_2	B_1	B_0		B[7] ~ LUT_EN 0: During reset, auto load LUT from OTP if it is valid. When PON/DRF, analog setting will follow the content of OTP. (Default) 1: During reset, no action to load LUT from OTP. When PON/DRF, analog setting will follow the content of MCU

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R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description
												B[6:5] ~ FOPT[1:0] FOPT function 00b: NO effect. Scan 1 frame after waveform finished (Default) 01b: Will scan 1 frame same as last output (of LUT) after waveform finished. 10b: Will scan 1 frame with Floating output after waveform finished. 11b: No Scan after waveform finished (Power off floating) B[4] ~ VCMZ VCOM Hi-Z option 0: NO effect. (default) 1: VCOM is always floating. B[3] ~ TS_AUTO Temperature Sensor auto option 0: NO effect. 1: Before loading OTP, Temperature Sensor will be activated automatically one time. (default) B[2] ~ TIEG VGL power off option 0: NO effect. (default) 1: After power off booster, VGL will be tied to GND. B[1] ~ NORG VCOM display end GND option 0: NO effect. (default) 1: After refreshing display, VCOM is tied to GND before power off B[0] ~ VC_LUTZ VCOM display end floating option 0: NO effect. 1: After refreshing display, the output of VCOM is set to floating automatically (default) VCMZ, NORG, FOPT, VC_LUTZ and TIEG detail setting refer to appendix.

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<table>
<tr><th>R/W#</th><th>D/C#</th><th>Hex</th><th>D7</th><th>D6</th><th>D5</th><th>D4</th><th>D3</th><th>D2</th><th>D1</th><th>D0</th><th>Command</th><th>Description</th></tr>
<tr><td>0</td><td>0</td><td>01</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>PWR</td><td>Power setting Register
A[5:0] = 0Fh [POR]
B[1:0] = 0h [POR]
C[6:0] = 00h [POR]
D[6:0] = 00h [POR]
E[6:0] = 00h [POR]
F[6:0] = 00h [POR]</td></tr>
<tr><td>0</td><td>1</td><td></td><td>0</td><td>0</td><td>1</td><td>1</td><td>A_3</td><td>A_2</td><td>A_1</td><td>A_0</td><td></td><td>A[5:4] needs to set as 11

A[3] ~ VSC2_EN
Source LV2 power selection:
0: External source power for VSH_LV2 and VSL_LV2 pin.
1: Internal DCDC function for generate source power. (default)
A[2] ~ VSC_EN
Source LV power selection:
0: External source power for VSH_LV and VSL_LV pin.
1: Internal DCDC function for generate source power. (default)
A[1] ~ VS_EN
Source LV power selection:
0: External source power for VSH and VSL pin.
1: Internal DCDC function for generate source power. (default)
A[0] ~ VG_EN
Gate power selection:
0: External gate power for VGH and VGL pin.
1: Internal DCDC function for generate gate power. (default)</td></tr>
<tr><td>0</td><td>1</td><td></td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>B_1</td><td>B_0</td><td></td><td>B[1:0] ~ VGPN [1:0]
Internal VGH / VGL Voltage Level Selection:
<table>
<tr><th>VGPN [1:0]</th><th>Gate Voltage Level</th></tr>
<tr><td>00</td><td>VGH=20V,VGL=-20V (Default)
VSH=15V, VSL=-15V</td></tr>
<tr><td>01</td><td>VGH=17V ,VGL=-17V
VSH=15V, VSL=-15V</td></tr>
<tr><td>10</td><td>VGH=13V,VGL=-13V
VSH=11V, VSL=-11V</td></tr>
<tr><td>11</td><td>VGH=10V,VGL=-10V
VSH=8V, VSL=-8V</td></tr>
</table></td></tr>
<tr><td>0</td><td>1</td><td></td><td>0</td><td>C_6</td><td>C_5</td><td>C_4</td><td>C_3</td><td>C_2</td><td>C_1</td><td>C_0</td><td></td><td>C[6:0] ~ VSPL [6:0]
Internal VSH_LV Voltage Level Selection:
<table>
<tr><th>VSPL[6:0]</th><th>VSH_LV Voltage Level</th></tr>
<tr><td>00h</td><td>3.0V (default)</td></tr>
<tr><td>01h</td><td>3.1V</td></tr>
<tr><td>02h</td><td>3.2V</td></tr>
<tr><td>03h</td><td>3.3V</td></tr>
<tr><td>:</td><td>:</td></tr>
<tr><td>77h</td><td>14.9V</td></tr>
<tr><td>78h</td><td>15.0V</td></tr>
<tr><td>Other</td><td>Reserved</td></tr>
</table>
If VSPL>= VGH-2V,
VSPL setting will be remapped as VGH-2V</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
</table>

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<table>
<tr><th>R/W#</th><th>D/C#</th><th>Hex</th><th>D7</th><th>D6</th><th>D5</th><th>D4</th><th>D3</th><th>D2</th><th>D1</th><th>D0</th><th>Command</th><th>Description</th></tr>
<tr><td>0</td><td>1</td><td></td><td>0</td><td>D_6</td><td>D_5</td><td>D_4</td><td>D_3</td><td>D_2</td><td>D_1</td><td>D_0</td><td></td><td>D[6:0] ~ VSNL[6:0]
Internal VSL_LV Voltage Level Selection:
<table><tr><td>VSNL[6:0]</td><td>VSL_LV Voltage Level</td></tr><tr><td>00h</td><td>-3.0V (default)</td></tr><tr><td>01h</td><td>-3.1V</td></tr><tr><td>02h</td><td>-3.2V</td></tr><tr><td>03h</td><td>-3.3V</td></tr><tr><td>:</td><td>:</td></tr><tr><td>77h</td><td>-14.9V</td></tr><tr><td>78h</td><td>-15.0V</td></tr><tr><td>Other</td><td>Reserved</td></tr></table>
If VSNL<= VGL+2V,
VSNL setting will be remapped as VGL+2V</td></tr>
<tr><td>0</td><td>1</td><td></td><td>0</td><td>E_6</td><td>E_5</td><td>E_4</td><td>E_3</td><td>E_2</td><td>E_1</td><td>E_0</td><td></td><td>E[6:0] ~ VSPL2[6:0]
Internal VSH_LV2 Voltage Level Selection:
<table><tr><td>VSPL2[6:0]</td><td>VSH_LV2 Voltage Level</td></tr><tr><td>00h</td><td>3.0V (default)</td></tr><tr><td>01h</td><td>3.1V</td></tr><tr><td>02h</td><td>3.2V</td></tr><tr><td>03h</td><td>3.3V</td></tr><tr><td>:</td><td>:</td></tr><tr><td>77h</td><td>14.9V</td></tr><tr><td>78h</td><td>15.0V</td></tr><tr><td>Other</td><td>Reserved</td></tr></table>
If VSPL2>= VGH-2V,
VSPL2 setting will be remapped as VGH-2V</td></tr>
<tr><td>0</td><td>1</td><td></td><td>0</td><td>F_6</td><td>F_5</td><td>F_4</td><td>F_3</td><td>F_2</td><td>F_1</td><td>F_0</td><td></td><td>F[6:0] ~ VSNL2[6:0]
Internal VSL_LV2 Voltage Level Selection:
<table><tr><td>VSNL2[6:0]</td><td>VSL_LV 2Voltage Level</td></tr><tr><td>00h</td><td>-3.0V (default)</td></tr><tr><td>01h</td><td>-3.1V</td></tr><tr><td>02h</td><td>-3.2V</td></tr><tr><td>03h</td><td>-3.3V</td></tr><tr><td>:</td><td>:</td></tr><tr><td>77h</td><td>-14.9V</td></tr><tr><td>78h</td><td>-15.0V</td></tr><tr><td>Other</td><td>Reserved</td></tr></table>
If VSNL2<= VGL+2V,
VSNL2 setting will be remapped as VGL+2V</td></tr>
<tr><td colspan="13"></td></tr>
<tr><td>0</td><td>0</td><td>02</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>POF</td><td>Power OFF Command Register</td></tr>
<tr><td>0</td><td>1</td><td></td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td></td><td>After power off command, driver will power off based on the Power OFF Sequence, then BUSY_N signal will become “0”.
The Power OFF command will turn off DCDC, source driver, gate driver, VCOM driver, temperature sensor, but register and SRAM data will keep until VDD off.
SD output will base on previous condition.
*Remark: POF works at PON only</td></tr>
</table>

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R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description
0	0	03	0	0	0	0	0	0	1	1	POFS	Power on/off Sequence Setting Register A[7:0] = 00h [POR]
0	1		A_7	A_6	A_5	A_4	A_3	A_2	A_1	A_0		A[5:4] ~ T_VDPG_OFF[1:0] Power off delay from VGL to VGH 00b: 20ms (default) 01b: 40ms 10b: 60ms 11b: 80ms A[3:2] ~ T_VDS_ON[1:0] Power on delay from VGH to VSHX/VSLX 00b: 2ms (default) 01b: 4ms 10b: 6ms 11b: 8ms A[1:0] ~ T_VDS_OFF[1:0] Power off delay from VSHX/VSLX to VGH 00b: 20ms (default) 01b: 40ms 10b: 60ms 11b : 80ms
0	1		0	1	0	1	0	1	0	0		
0	1		0	0	0	0	0	0	0	0		
0	1		0	1	0	0	0	1	0	0		
0	0	04	0	0	0	0	0	1	0	0	PON	Power ON Command Register After the Power ON command, driver will power on based on the Power ON Sequence. After power on command and all power sequence are ready, then BUSY_N signal will become “1”. * Remark: PON Include booster on, VSHx/VSLx regulator on With default BTST, timing is >80ms

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<table>
<tr><th>R/W#</th><th>D/C#</th><th>Hex</th><th>D7</th><th>D6</th><th>D5</th><th>D4</th><th>D3</th><th>D2</th><th>D1</th><th>D0</th><th>Command</th><th>Description</th></tr>
<tr><td>0</td><td>0</td><td>05</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td><td>BTST1</td><td>VPC Booster Soft Start Setting Register
A[6:0] = 00h [POR]
B[6:0] = 1Fh [POR]
C[6:0] = 1Fh [POR]
D[6:0] = 2Ch [POR]</td></tr>
<tr><td>0</td><td>1</td><td></td><td>0</td><td>A_6</td><td>A_5</td><td>A_4</td><td>A_3</td><td>A_2</td><td>A_1</td><td>A_0</td><td></td><td rowspan="4">A[6:4] ~ VPCSSC_ONT [2:0]
VPC booster maximum MOSFET ON time
(reserved)
A[6:4] = 000 (Default)

A[3:2] ~ T_VPCSSA [1:0]
VPC booster soft start Phase A duration
A[1:0] ~ T_VPCSSB [1:0]
VPC booster soft start Phase B duration
<table><tr><td></td><td>Soft Start Phase Period (ms)</td></tr><tr><td>00</td><td>4 (Default)</td></tr><tr><td>01</td><td>8</td></tr><tr><td>10</td><td>12</td></tr><tr><td>11</td><td>16</td></tr></table>
B[6:4] ~ VPCSSA_DRV [2:0], = 001 (Default)
VPC Phase A Driving Strength
C[6:4] ~ VPCSSB_DRV [2:0], = 001 (Default)
VPC Phase B Driving Strength
D[6:4] ~ VPCSSC_DRV [2:0] = 010 (Default)
VPC Phase C Driving Strength
<table><tr><td colspan="4">Driving Strength</td></tr><tr><td>000</td><td>(reserved)</td><td>100</td><td>3</td></tr><tr><td>001</td><td>0</td><td>101</td><td>4</td></tr><tr><td>010</td><td>1</td><td>110</td><td>5</td></tr><tr><td>011</td><td>2</td><td>111</td><td>6(strongest)</td></tr></table>
B[3:0] ~ VPCSSA_OFFT[[3:0], = 1111 (Default)
VPC Phase A Minimum OFF Time
C[3:0] ~ VPCSSB_OFFT[[3:0], = 1111 (Default)
VPC Phase B Minimum OFF Time
D[3:0] ~ VPCSSC_OFFT[[3:0], = 1100 (Default)
VPC Phase C Minimum OFF Time
<table><tr><td colspan="4">Minimum OFF Time (setting)</td></tr><tr><td>0000</td><td>OFFC0</td><td>1000</td><td>OFFC8</td></tr><tr><td>0001</td><td>OFFC1</td><td>1001</td><td>OFFC9</td></tr><tr><td>0010</td><td>OFFC2</td><td>1010</td><td>OFFCA</td></tr><tr><td>0011</td><td>OFFC3</td><td>1011</td><td>OFFCB</td></tr><tr><td>0100</td><td>OFFC4</td><td>1100</td><td>OFFCC</td></tr><tr><td>0101</td><td>OFFC5</td><td>1101</td><td>OFFCD</td></tr><tr><td>0110</td><td>OFFC6</td><td>1110</td><td>OFFCE</td></tr><tr><td>0111</td><td>OFFC7</td><td>1111</td><td>OFFCF</td></tr></table></td></tr>
<tr><td>0</td><td>1</td><td></td><td>0</td><td>B_6</td><td>B_5</td><td>B_4</td><td>B_3</td><td>B_2</td><td>B_1</td><td>B_0</td><td></td></tr>
<tr><td>0</td><td>1</td><td></td><td>0</td><td>C_6</td><td>C_5</td><td>C_4</td><td>C_3</td><td>C_2</td><td>C_1</td><td>C_0</td><td></td></tr>
<tr><td>0</td><td>1</td><td></td><td>0</td><td>D_6</td><td>D_5</td><td>D_4</td><td>D_3</td><td>D_2</td><td>D_1</td><td>D_0</td><td></td></tr>
</table>

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<table>
<tr><th>R/W#</th><th>D/C#</th><th>Hex</th><th>D7</th><th>D6</th><th>D5</th><th>D4</th><th>D3</th><th>D2</th><th>D1</th><th>D0</th><th>Command</th><th>Description</th></tr>
<tr><td>0</td><td>0</td><td>06</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>BTST2</td><td>VPH Booster Soft Start Setting Register (for VGH)
A[6:0] = 00h [POR]
B[6:0] = 1Fh [POR]
C[6:0] = 1Fh [POR]
D[6:0] = 45h [POR]</td></tr>
<tr><td>0</td><td>1</td><td></td><td>0</td><td>A_6</td><td>A_5</td><td>A_4</td><td>A_3</td><td>A_2</td><td>A_1</td><td>A_0</td><td></td><td rowspan="4">A[6:4] ~ VPHSSC_ONT [2:0]
VPH booster maximum MOSFET ON time
(reserved)
A[6:4] = 000 (Default)

A[3:2] ~ T_VPHSSA [1:0]
VPH booster soft start Phase A duration
A[1:0] ~ T_VPHSSB [1:0]
VPH booster soft start Phase B duration
<table><tr><td></td><td>Soft Start Phase Period (ms)</td></tr><tr><td>00</td><td>10 (Default)</td></tr><tr><td>01</td><td>20</td></tr><tr><td>10</td><td>30</td></tr><tr><td>11</td><td>40</td></tr></table>
B[6:4] ~ VPHSSA_DRV [2:0], = 001 (Default)
VPH Phase A Driving Strength
C[6:4] ~ VPHSSB_DRV [2:0], = 001 (Default)
VPH Phase B Driving Strength
D[6:4] ~ VPHSSC_DRV [2:0] = 011 (Default)
VPH Phase C Driving Strength
<table><tr><td colspan="4">Driving Strength</td></tr><tr><td>000</td><td>(reserved)</td><td>100</td><td>3</td></tr><tr><td>001</td><td>0</td><td>101</td><td>4</td></tr><tr><td>010</td><td>1</td><td>110</td><td>5</td></tr><tr><td>011</td><td>2</td><td>111</td><td>6(strongest)</td></tr></table>
B[3:0] ~ VPHSSA_OFFT[[3:0], = 1111 (Default)
VPH Phase A Minimum OFF Time
C[3:0] ~ VPHSSB_OFFT[[3:0], = 1111 (Default)
VPH Phase B Minimum OFF Time
D[3:0] ~ VPHSSC_OFFT[[3:0], = 0101 (Default)
VPH Phase C Minimum OFF Time
<table><tr><td colspan="4">Minimum OFF Time (setting)</td></tr><tr><td>0000</td><td>OFFH0</td><td>1000</td><td>OFFH8</td></tr><tr><td>0001</td><td>OFFH1</td><td>1001</td><td>OFFH9</td></tr><tr><td>0010</td><td>OFFH2</td><td>1010</td><td>OFFHA</td></tr><tr><td>0011</td><td>OFFH3</td><td>1011</td><td>OFFHB</td></tr><tr><td>0100</td><td>OFFH4</td><td>1100</td><td>OFFHC</td></tr><tr><td>0101</td><td>OFFH5</td><td>1101</td><td>OFFHD</td></tr><tr><td>0110</td><td>OFFH6</td><td>1110</td><td>OFFHE</td></tr><tr><td>0111</td><td>OFFH7</td><td>1111</td><td>OFFHF</td></tr></table></td></tr>
<tr><td>0</td><td>1</td><td></td><td>0</td><td>B_6</td><td>B_5</td><td>B_4</td><td>B_3</td><td>B_2</td><td>B_1</td><td>B_0</td><td></td></tr>
<tr><td>0</td><td>1</td><td></td><td>0</td><td>C_6</td><td>C_5</td><td>C_4</td><td>C_3</td><td>C_2</td><td>C_1</td><td>C_0</td><td></td></tr>
<tr><td>0</td><td>1</td><td></td><td>0</td><td>D_6</td><td>D_5</td><td>D_4</td><td>D_3</td><td>D_2</td><td>D_1</td><td>D_0</td><td></td></tr>
<tr><td colspan="13"></td></tr>
<tr><td>0</td><td>0</td><td>07</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>DSLP</td><td rowspan="2">Deep Sleep Register
This command makes the chip enter the deep-sleep mode. The deep sleep mode could return to stand-by mode by hardware reset assertion.
The only one parameter is a check code, the command would be executed if check code is A5h.</td></tr>
<tr><td>0</td><td>1</td><td></td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td><td></td></tr>
</table>

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R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description
0	0	10	0	0	0	1	0	0	0	0	DTM	Data Start transmission Register This command indicates that user starts to transmit data. Then write to SRAM. While complete data transmission, user must send a Data Refresh command (R12H). Then the chip will start to send data/VCOM for panel. Pixel Mode = 0: 2-bit/pixel KPixel[1:0] Source Driver Output
3 bit per pixel												
0	1		0	KPixel1[2:0]			0	KPixel2[2:0]				
:				:				:				
0	1		0	KPixel(2M-1)[2:0]			0	KPixel(2M)[2:0]				
2 bit per pixel												
0	1		KPixel1[1:0]		KPixel2[1:0]		KPixel3[1:0]		KPixel4[1:0]			
:												
0	1		KPixel(4M-3)[1:0]		KPixel(4M-2)[1:0]		KPixel(4M-1)[1:0]		KPixel(4M)[1:0]			

	DDX=1 (Default)	DDX=0
00b	Gray 0	Gray 3
01b	Gray 1	Gray 2
10b	Gray 2	Gray 1
11b	Gray 3	Gray 0

Pixel Mode = 1: 3-bit/pixel
KPixel[2:0] Source Driver Output

	DDX=1 (Default)	DDX=0
000b	Gray 0	Gray 7
001b	Gray 1	Gray 6
010b	Gray 2	Gray 5
011b	Gray 3	Gray 4
100b	Gray 4	Gray 3
101b	Gray 5	Gray 2
110b	Gray 6	Gray 1
111b	Gray 7	Gray 0

After issue this command, the host must send at least 1 byte data to the device.

0	0	12	0	0	0	1	0	0	1	0	DRF	Display Refresh Command Register After this command is issued, driver will refresh display (data/FPL_VCOM) according to SRAM data and LUT. A[0] ~ AC_DC_VCOM AC / DC VCOM selection 0: ACVCOM, FPL_VCOM will follow LUTC when updating image (default) 1: DCVCOM, FPL_VCOM will always be VCOMDC when updating image After Display Refresh command, BUSY_N signal will become “0” until display update is finished.
0	1		0	0	0	0	0	0	0	A_0		
0	0	17	0	0	0	1	0	1	1	1	AUTO	Auto Sequence Register
0	1		A_7	A_6	A_5	A_4	A_3	A_2	A_1	A_0		This command makes the chip enter the auto sequence Single-chip application ONLY. Auto Sequence Option 0xA5: Start Auto Sequence (PON > DRF > POF) 0xA7: Start Auto Sequence (PON > DRF > POF > DSLP). Others: No effect BUSY_N signal will become “0” until Auto Sequence is finished.

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<table>
<tr><th>R/W#</th><th>D/C#</th><th>Hex</th><th>D7</th><th>D6</th><th>D5</th><th>D4</th><th>D3</th><th>D2</th><th>D1</th><th>D0</th><th>Command</th><th>Description</th></tr>
<tr><td>0</td><td>0</td><td>30</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>PLL</td><td>Frame Rate Control register
The command controls the clock frequency.
A[2:0] = 2h [POR]</td></tr>
<tr><td>0</td><td>1</td><td></td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>A_2</td><td>A_1</td><td>A_0</td><td></td><td>It supports the following frame rates.
A[2:0] ~ FR[2:0]
Frame Rate Selection

<table><tr><td>FR[2:0]</td><td>Frame Rate Selection</td></tr><tr><td>000</td><td>25Hz</td></tr><tr><td>001</td><td>32Hz</td></tr><tr><td>010</td><td>43Hz (Default)</td></tr><tr><td>011</td><td>50Hz</td></tr><tr><td>100</td><td>65Hz</td></tr><tr><td>101</td><td>75Hz</td></tr><tr><td>110</td><td>85Hz</td></tr><tr><td>111</td><td>100Hz</td></tr></table>
Detail refer to Appendix</td></tr>
<tr><td colspan="13"></td></tr>
<tr><td>0</td><td>0</td><td>40</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>TSC</td><td>Temperature Sensor Command Register
This command enables internal or external temperature sensor.
BUSY_N will go low during temperature sensor is under operation. Then the temperature value can be read.

When TSE (R41h) is set to 0, this command reads internal temperature sensor value. 0.5degC step
A[11:3] ~ TS [8:0]
A [2:0] = 000
<table><tr><td>A [11:0]</td><td>Return Value(degC)</td></tr><tr><td>E70h</td><td>-25</td></tr><tr><td>…</td><td>…</td></tr><tr><td>FF0h</td><td>-1</td></tr><tr><td>FF8h</td><td>-0.5</td></tr><tr><td>000h</td><td>0</td></tr><tr><td>008h</td><td>0.5</td></tr><tr><td>010h</td><td>1</td></tr><tr><td>…</td><td>…</td></tr><tr><td>190h</td><td>25</td></tr><tr><td>…</td><td>…</td></tr><tr><td>3B0h</td><td>59</td></tr><tr><td>3B8h</td><td>59.5</td></tr><tr><td>3C0h</td><td>60</td></tr></table>
When TSE (R41h) is set to 1, this command reads external temperature sensor value. A[11:0] ~ Return the value refer to Session 6.6 External Temperature Sensor I2C Single Master Interface</td></tr>
<tr><td>1</td><td>1</td><td></td><td>A_{11}</td><td>A_{10}</td><td>A_9</td><td>A_8</td><td>A_7</td><td>A_6</td><td>A_5</td><td>A_4</td><td></td><td></td></tr>
<tr><td>1</td><td>1</td><td></td><td>A_3</td><td>A_2</td><td>A_1</td><td>A_0</td><td>0</td><td>0</td><td>0</td><td>0</td><td></td><td></td></tr>
</table>

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<table>
<tr><th>R/W#</th><th>D/C#</th><th>Hex</th><th>D7</th><th>D6</th><th>D5</th><th>D4</th><th>D3</th><th>D2</th><th>D1</th><th>D0</th><th>Command</th><th>Description</th></tr>
<tr><td>0</td><td>0</td><td>41</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>TSE</td><td>Temperature Sensor Enable Register
This command selects Internal or External temperature sensor.
A[7:0] = 00h [POR]</td></tr>
<tr><td>0</td><td>1</td><td></td><td>A_7</td><td>0</td><td>0</td><td>0</td><td>A_3</td><td>A_2</td><td>A_1</td><td>A_0</td><td></td><td>A[7] ~ TSE
Internal temperature sensor switch:
0: Select internal temperature sensor (default)
1: Select external temperature sensor.

A[3:0] ~ TO[3:0]
Temperature offset:
<table>
<tr><th>TO[3:0]</th><th>Calculation</th><th>TO[3:0]</th><th>Calculation</th></tr>
<tr><td>0000</td><td>+0</td><td>1000</td><td>-4</td></tr>
<tr><td>0001</td><td>+0.5</td><td>1001</td><td>-3.5</td></tr>
<tr><td>0010</td><td>+1</td><td>1010</td><td>-3</td></tr>
<tr><td>0011</td><td>+1.5</td><td>1011</td><td>-2.5</td></tr>
<tr><td>0100</td><td>+2</td><td>1100</td><td>-2</td></tr>
<tr><td>0101</td><td>+2.5</td><td>1101</td><td>-2.5</td></tr>
<tr><td>0110</td><td>+3</td><td>1110</td><td>-1</td></tr>
<tr><td>0111</td><td>+3.5</td><td>1111</td><td>-0.5</td></tr>
</table></td></tr>
<tr><td colspan="13"></td></tr>
<tr><td>0</td><td>0</td><td>42</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>TSW</td><td>Temperature Sensor Write Register
This command could write data to the external temperature sensor.
A[7:0] = 00h [POR]
B[7:0] = 00h [POR]
C[7:0] = 00h [POR]</td></tr>
<tr><td>0</td><td>1</td><td></td><td>A_7</td><td>A_6</td><td>A_5</td><td>A_4</td><td>A_3</td><td>A_2</td><td>A_1</td><td>A_0</td><td></td><td rowspan="3">A[7:0] ~ WATTR [7:0]
WATTR [7:6]: I2C Write Byte Number
00: 1 byte (head byte only)
01: 2 bytes (head byte + pointer)
10: 3 bytes (head byte + pointer + 1st parameter)
11: 4 bytes (head byte + pointer + 1st parameter + 2nd parameter)

WATTR [5:3]: User-defined address bits (A2, A1, A0)

WATTR [2:0]: Pointer setting

B[7:0] ~ WMSB[7:0]:
1st parameter of write-data to external temperature sensor

C[7:0] ~ WLSB[7:0]:
2nd parameter of write-data to external temperature sensor

BUSY_N signal will become “0” until external Temperature Sensor write is finished.</td></tr>
<tr><td>0</td><td>1</td><td></td><td>B_7</td><td>B_6</td><td>B_5</td><td>B_4</td><td>B_3</td><td>B_2</td><td>B_1</td><td>B_0</td><td></td></tr>
<tr><td>0</td><td>1</td><td></td><td>C_7</td><td>C_6</td><td>C_5</td><td>C_4</td><td>C_3</td><td>C_2</td><td>C_1</td><td>C_0</td><td></td></tr>
</table>

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<table>
<tr><th>R/W#</th><th>D/C#</th><th>Hex</th><th>D7</th><th>D6</th><th>D5</th><th>D4</th><th>D3</th><th>D2</th><th>D1</th><th>D0</th><th>Command</th><th>Description</th></tr>
<tr><td>0</td><td>0</td><td>43</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>TSR</td><td>Temperature Sensor Read Register
This command read data from the external temperature sensor
A[7:0] = RMSB[7:0]: MSByte of read-data from external temperature sensor
B[7:0] = RLSB[7:0] : LSByte of read-data from external temperature sensor</td></tr>
<tr><td>1</td><td>1</td><td></td><td>A_7</td><td>A_6</td><td>A_5</td><td>A_4</td><td>A_3</td><td>A_2</td><td>A_1</td><td>A_0</td><td></td><td></td></tr>
<tr><td>1</td><td>1</td><td></td><td>B_7</td><td>B_6</td><td>B_5</td><td>B_4</td><td>B_3</td><td>B_2</td><td>B_1</td><td>B_0</td><td></td><td></td></tr>
<tr><td colspan="13"></td></tr>
<tr><td>0</td><td>0</td><td>50</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>CDI</td><td>VCOM and DATA interval setting Register
This command indicates the Border Output Selection and the interval between Vcom and data output
A[7:0] = 9Eh [POR]</td></tr>
<tr><td>0</td><td>1</td><td></td><td>A_7</td><td>A_6</td><td>A_5</td><td>A_4</td><td>A_3</td><td>A_2</td><td>A_1</td><td>A_0</td><td></td><td>A[7:5]~VBD [2:0]
Border Output Selection:
<table>
<tr><td></td><td>DDX=1</td><td>DDX=0</td></tr>
<tr><td>VBD[2:0]</td><td>LUT (Default)</td><td>LUT</td></tr>
<tr><td>000</td><td>Gray 0</td><td>HIZ</td></tr>
<tr><td>001</td><td>Gray 1</td><td>Gray 6</td></tr>
<tr><td>010</td><td>Gray 2</td><td>Gray 5</td></tr>
<tr><td>011</td><td>Gray 3</td><td>Gray 4</td></tr>
<tr><td>100</td><td>Gray 4 (Default)</td><td>Gray 3</td></tr>
<tr><td>101</td><td>Gray 5</td><td>Gray 2</td></tr>
<tr><td>110</td><td>Gray 6</td><td>Gray 1</td></tr>
<tr><td>111</td><td>HIZ</td><td>Gray 0</td></tr>
</table>
A[4]: DDX
Data Polarity for Pixel Data (See DTM command)
0: Invert
1: Non-invert (Default)

VBD will output as VCOMBD, which output the target source level with DCVCOM offset.
e.g. Gray 1 waveform is mapping to 15V, DCVCOM=-2V. VCOMBD=15V+(-2V)=13V.

A[3:0] ~ CDI[3:0]
Vcom and Data Interval (time unit)
<table>
<tr><td>CDI [3:0]</td><td>Interval</td><td>CDI [3:0]</td><td>Interval</td></tr>
<tr><td>0h</td><td>17</td><td>8h</td><td>9</td></tr>
<tr><td>1h</td><td>16</td><td>9h</td><td>8</td></tr>
<tr><td>2h</td><td>15</td><td>Ah</td><td>7</td></tr>
<tr><td>3h</td><td>14</td><td>Bh</td><td>6</td></tr>
<tr><td>4h</td><td>13</td><td>Ch</td><td>5</td></tr>
<tr><td>5h</td><td>12</td><td>Dh</td><td>4</td></tr>
<tr><td>6h</td><td>11</td><td>Eh</td><td>3 (Default)</td></tr>
<tr><td>7h</td><td>10</td><td>Fh</td><td>2</td></tr>
</table></td></tr>
<tr><td colspan="13"></td></tr>
<tr><td>0</td><td>0</td><td>51</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>LPD</td><td>Lower Power Detection Register
This command indicates the input power condition.
Host can read this flag to learn the battery condition.</td></tr>
<tr><td>1</td><td>1</td><td></td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>A_0</td><td></td><td>BUSY_N will go low during input power condition is under operation. Then the LPD status can be read.
A[0] ~ LPD status
0: Low power input
(VDD<2.5V, selected by LVD_SEL[1:0] in command LVSEL)
1: Normal (default)</td></tr>
<tr><td colspan="13"></td></tr>
</table>

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<table>
<tr><th>R/W#</th><th>D/C#</th><th>Hex</th><th>D7</th><th>D6</th><th>D5</th><th>D4</th><th>D3</th><th>D2</th><th>D1</th><th>D0</th><th>Command</th><th>Description</th></tr>
<tr><td>0</td><td>0</td><td>61</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>TRES</td><td>Resolution setting Register
This command defines alternative resolution</td></tr>
<tr><td>0</td><td>1</td><td></td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>A_9</td><td>A_8</td><td rowspan="5">A[9:0] ~ HRES[9:0]
Horizontal Display Resolution
Remark: Horizontal resolution should be 4-multiple.
B[9:0] ~ VRES[9:0]
Vertical Display Resolution

e.g. HRES= 258h, VRES= 1C0h
<table><tr><td>UD,SHL</td><td>Source and gate sequence</td></tr><tr><td>00</td><td>S599,G447 to S0,G0</td></tr><tr><td>01</td><td>S0,G447 to S599,G0</td></tr><tr><td>10</td><td>S599,G0 to S0,G447</td></tr><tr><td>11</td><td>S0,G0 to S599,G447</td></tr></table>
Remark:
1) Both PSR.RES & TRES command can set panel resolution. Priority will be given to the last received PSR or TRES command.
2) VRES[8:0] >= 240</td></tr>
<tr><td>0</td><td>1</td><td></td><td>A_7</td><td>A_6</td><td>A_5</td><td>A_4</td><td>A_3</td><td>A_2</td><td>A_1</td><td>A_0</td></tr>
<tr><td>0</td><td>1</td><td></td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>B_9</td><td>B_8</td></tr>
<tr><td>0</td><td>1</td><td></td><td>B_7</td><td>B_6</td><td>B_5</td><td>B_4</td><td>B_3</td><td>B_2</td><td>B_1</td><td>B_0</td></tr>
<tr><td colspan="13"></td></tr>
<tr><td>0</td><td>0</td><td>70</td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>REV</td><td>Chip Revision Register
The command is to read the ID
A[7:0] = 01h [POR]
B[7:0] ~ EINK Internal number = 01h [POR]
C[7:0] ~ IC version = 04h [POR]</td></tr>
<tr><td>1</td><td>1</td><td></td><td>A_7</td><td>A_6</td><td>A_5</td><td>A_4</td><td>A_3</td><td>A_2</td><td>A_1</td><td>A_0</td></tr>
<tr><td>1</td><td>1</td><td></td><td>B_7</td><td>B_6</td><td>B_5</td><td>B_4</td><td>B_3</td><td>B_2</td><td>B_1</td><td>B_0</td></tr>
<tr><td>1</td><td>1</td><td></td><td>C_7</td><td>C_6</td><td>C_5</td><td>C_4</td><td>C_3</td><td>C_2</td><td>C_1</td><td>C_0</td></tr>
</table>

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R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description
0	0	80	1	0	0	0	0	0	0	0	AMV	Auto Measurement VCOM Register This command implements related VCOM sensing setting. A[7:0] = 00h [POR] B[7:0] = 00h [POR]
0	1		A_7	A_6	A_5	A_4	A_3	A_2	A_1	A_0		A[7:6] ~ P[1:0] Number of sensing Points 00: 2 (Default) 01: 4 10: 8 11: 16 A[5:4] ~AMVT[1:0] Auto Measure Vcom Time: Sensing Time 00: 5 sec. (Default) 01: 10 sec. 10: 15 sec. 11: 20 sec. A[3] ~ AMVX VCOM measurement Gate settings 0: Gate scan normally during VCOM measurement (Default) 1: All Gates ON during VCOM measurement A[2] ~ AMVS Source output of AMV: 0: Set Source output to 0V during Auto Measure VCOM period. (Default) 1: Set Source output to VSH_LV during Auto Measure VCOM period. A[1] ~ AMV Analogy Option 0: Measure VCOM and update value to VDCS[6:0] 1: Gate scan only. Measure VCOM externally by probing the VCOM pad. A[0] ~ AMVE Auto Measure Vcom Enable (/Disable): 0: Disabled (Default) 1: Enabled B[1:0] needs to set as 10 Requirement: 1) AMV works at PON only 2) BUSY_N signal will become “0” until Vcom sensing is finished.
0	1		0	0	0	0	0	0	1	0		

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<table>
<tr><th>R/W#</th><th>D/C#</th><th>Hex</th><th>D7</th><th>D6</th><th>D5</th><th>D4</th><th>D3</th><th>D2</th><th>D1</th><th>D0</th><th>Command</th><th>Description</th></tr>
<tr><td>0</td><td>0</td><td>81</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>VV</td><td>Auto Measurement VCOM Register
This command gets the Vcom value after AMV.</td></tr>
<tr><td>1</td><td>1</td><td></td><td>1</td><td>A_6</td><td>A_5</td><td>A_4</td><td>A_3</td><td>A_2</td><td>A_1</td><td>A_0</td><td></td><td>A[6:0] ~ VV[6:0]:
Vcom read Value
<table><tr><td>VV[6:0]</td><td>Vcom read value</td></tr><tr><td>00h</td><td>Reserved</td></tr><tr><td>01h</td><td>Reserved</td></tr><tr><td>02h</td><td>-0.10V</td></tr><tr><td>03h</td><td>-0.15V</td></tr><tr><td>04h</td><td>-0.20V</td></tr><tr><td>…</td><td>…</td></tr><tr><td>50h</td><td>-4.00V</td></tr><tr><td>others</td><td>-4.00V</td></tr></table></td></tr>
<tr><td colspan="13"></td></tr>
<tr><td>0</td><td>0</td><td>82</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>VDCS</td><td>VCM_DC Setting Register
This command sets VCOM_DC value.
A[7:0] = 00h [POR]</td></tr>
<tr><td>0</td><td>1</td><td></td><td>A_7</td><td>A_6</td><td>A_5</td><td>A_4</td><td>A_3</td><td>A_2</td><td>A_1</td><td>A_0</td><td></td><td>A[6:0] ~ VDCS[6:0]:
VCOM_DC Setting
<table><tr><td>VDCS [6:0]</td><td>VCOM_DC Setting</td></tr><tr><td>00h</td><td>Reserved</td></tr><tr><td>01h</td><td>Reserved</td></tr><tr><td>02h</td><td>-0.10V (Default)</td></tr><tr><td>03h</td><td>-0.15V</td></tr><tr><td>04h</td><td>-0.20V</td></tr><tr><td>…</td><td>…</td></tr><tr><td>50h</td><td>-4.00V</td></tr><tr><td>others</td><td>-4.00V</td></tr></table></td></tr>
<tr><td colspan="13"></td></tr>
<tr><td>0</td><td>0</td><td>83</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>PTLW</td><td rowspan="11">PARTIAL WINDOW
This command sets partial window address.
A[9:0] = 000h [POR]
B[9:0] = 31Fh [POR]
C[9:0] = 000h [POR]
D[9:0] = 257h [POR]
E[7:0] = 00h [POR]

A[9:0] ~ HRST [9:0]
Horizontal start address
B[9:0] ~ HRED [9:0]
Horizontal end address
C[9:0] ~ VRST [9:0]
Vertical start address
D[9:0] ~ VRED [9:0]
Vertical end address

E[0] ~ PMODE
0: disable partial mode (Default)
1: enable partial mode

Gate scan both inside and outside of the partial window

Requirement:
1) HRST [9:0]<= HRED [9:0]
2) VRST [9:0]<= VRED [9:0]</td></tr>
<tr><td>0</td><td>1</td><td></td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>A_9</td><td>A_8</td></tr>
<tr><td>0</td><td>1</td><td></td><td>A_7</td><td>A_6</td><td>A_5</td><td>A_4</td><td>A_3</td><td>A_2</td><td>A_1</td><td>A_0</td></tr>
<tr><td>0</td><td>1</td><td></td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>B_9</td><td>B_8</td></tr>
<tr><td>0</td><td>1</td><td></td><td>B_7</td><td>B_6</td><td>B_5</td><td>B_4</td><td>B_3</td><td>B_2</td><td>B_1</td><td>B_0</td></tr>
<tr><td>0</td><td>1</td><td></td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>C_9</td><td>C_8</td></tr>
<tr><td>0</td><td>1</td><td></td><td>C_7</td><td>C_6</td><td>C_5</td><td>C_4</td><td>C_3</td><td>C_2</td><td>C_1</td><td>C_0</td></tr>
<tr><td>0</td><td>1</td><td></td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>D_9</td><td>D_8</td></tr>
<tr><td>0</td><td>1</td><td></td><td>D_7</td><td>D_6</td><td>D_5</td><td>D_4</td><td>D_3</td><td>D_2</td><td>D_1</td><td>D_0</td></tr>
<tr><td>0</td><td>1</td><td></td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>E_0</td></tr>
<tr><td colspan="13"></td></tr>
</table>

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R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description
0	0	84	1	0	0	0	0	1	0	0	T_VDCS	TFT VCM Setting Register This command is set TFT_VCOM output A[7:0] = 00h [POR]
0	1		0	0	0	0	0	0	0	A_0		A[0] ~ T_VC_S TFT_VCOM output 0 – Follow VCOMDC (Default) 1 – 0V Remark: FPL_VCOM output is independent to T_VDCS
0	0	90	1	0	0	1	0	0	0	0	PGM	Program Mode This command is to set OTP program mode After this command is issued, the chip would enter the program mode. After the programming procedure completed, a hardware reset is necessary for leave the program mode BUSY_N signal will become 0 until PGM mode is ready.
0	0	91	1	0	0	1	0	0	0	1	APG	Active Program This command is to execute OTP program After this command is issued, the chip would program the OTP. BUSY_N signal will become 0 until the programming is completed. Requirement: In PON mode with internal programming power.
0	0	92	1	0	0	1	0	0	1	0	ROTP	Read OTP Data This command is to read the OTP content from SRAM. The 1st byte read is dummy byte. The 2nd byte read is the content of Address 0 in OTP The N+1th byte read is the content of Address n in OTP After issue this command, the host must read at least 1 byte data from the device.
1	1		1st ~ dummy 2nd ~ N+1th Parameter									

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<table>
<tr><th>R/W#</th><th>D/C#</th><th>Hex</th><th>D7</th><th>D6</th><th>D5</th><th>D4</th><th>D3</th><th>D2</th><th>D1</th><th>D0</th><th>Command</th><th>Description</th></tr>
<tr><td>0</td><td>0</td><td>E3</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>PWS</td><td>Power Saving Register
This command is sets for saving power VCOM/Source power saving during display refresh period.
A[7:0] = 75h [POR]</td></tr>
<tr><td>0</td><td>1</td><td></td><td>A_7</td><td>A_6</td><td>A_5</td><td>A_4</td><td>A_3</td><td>A_2</td><td>A_1</td><td>A_0</td><td></td><td>If the output voltage of VCOM/Source is from negative to positive or from positive to negative, the power saving mechanism will be activated. The active period width is defined by the following two parameters.

A[7:4] = VCOM_W [3:0]
<table><tr><th>VCOM_W [3:0]</th><th>VCOM_power saving width. (time unit)</th><th>VCOM_W [3:0]</th><th>VCOM_power saving width. (time unit)</th></tr><tr><td>0</td><td>Reserved</td><td>8</td><td>8</td></tr><tr><td>1</td><td>1</td><td>9</td><td>9</td></tr><tr><td>2</td><td>2</td><td>10</td><td>10</td></tr><tr><td>3</td><td>3</td><td>11</td><td>11</td></tr><tr><td>4</td><td>4</td><td>12</td><td>12</td></tr><tr><td>5</td><td>5</td><td>13</td><td>13</td></tr><tr><td>6</td><td>6</td><td>14</td><td>14</td></tr><tr><td>7</td><td>7 [Default]</td><td>15</td><td>15</td></tr></table>A[3:0] = SD_W [3:0]
<table><tr><th>SD_W [3:0]</th><th>Source power saving width [us]</th><th>SD_W [3:0]</th><th>Source power saving width [us]</th></tr><tr><td>0</td><td>Reserved</td><td>8</td><td>4</td></tr><tr><td>1</td><td>0.5</td><td>9</td><td>4.5</td></tr><tr><td>2</td><td>1</td><td>10</td><td>5</td></tr><tr><td>3</td><td>1.5</td><td>11</td><td>5.5</td></tr><tr><td>4</td><td>2</td><td>12</td><td>6</td></tr><tr><td>5</td><td>2.5[Default]</td><td>13</td><td>6.5</td></tr><tr><td>6</td><td>3</td><td>14</td><td>7</td></tr><tr><td>7</td><td>3.5</td><td>15</td><td>7.5</td></tr></table>Remark:
VCOM_W setting < CDI setting
SD_W setting < S2G setting</td></tr>
<tr><td colspan="13"></td></tr>
<tr><td>0</td><td>0</td><td>E4</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>LVSEL</td><td>LVD Voltage Select Register
This command is sets for low power detect level power
A[1:0] = 3h [POR]</td></tr>
<tr><td>0</td><td>1</td><td></td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>A_1</td><td>A_0</td><td></td><td>A[1:0] ~ LVD_SEL[1:0]
Low power detection threshold
00: 2.2V
01: 2.3V
10: 2.4V
11 : 2.5V (Default)</td></tr>
</table>

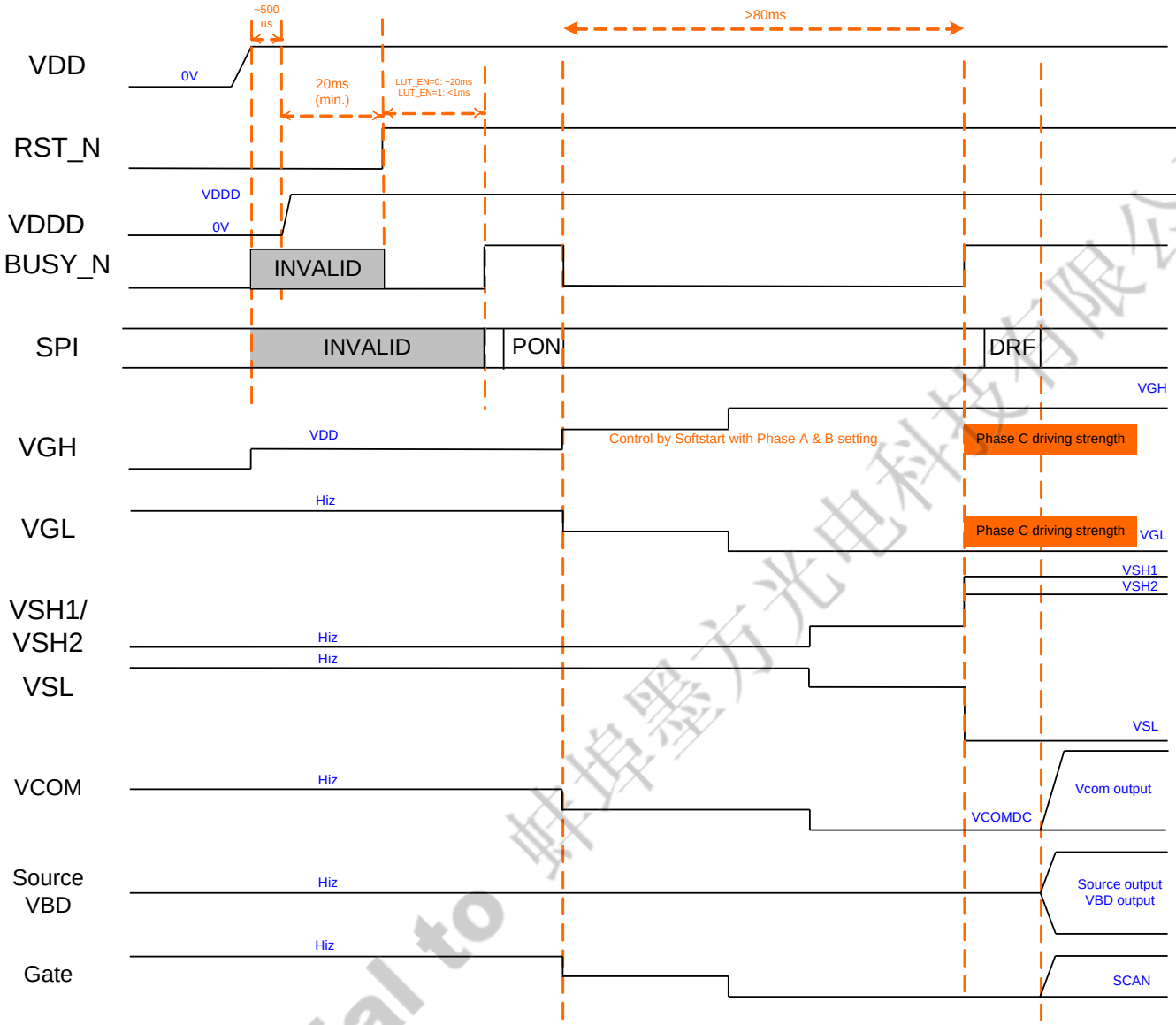
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R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description
0	0	E5	1	1	1	0	0	1	0	1	SPIM	Quad SPI Mode This command is set for Stardard 4 wire SPI interface A[1:0] = 0h [POR]
0	1		0	0	0	0	0	0	A_1	A_0		A[1:0] ~ SM[1:0] Dual / Quad SPI mode selection 00: Single Data SPI 01: Dual Data SPI 10: Quad Data SPI 11: Single Data SPI Remark: The Data byte only be affected DTM, ROTP commands

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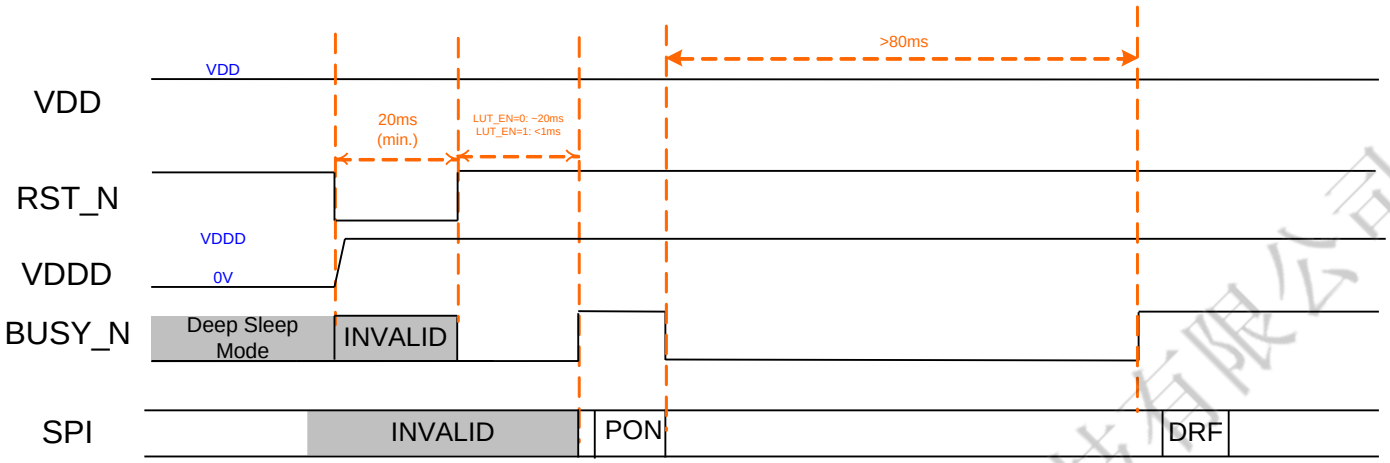
9 Display on/off Sequence

9.1 Display on sequence from VDD power on

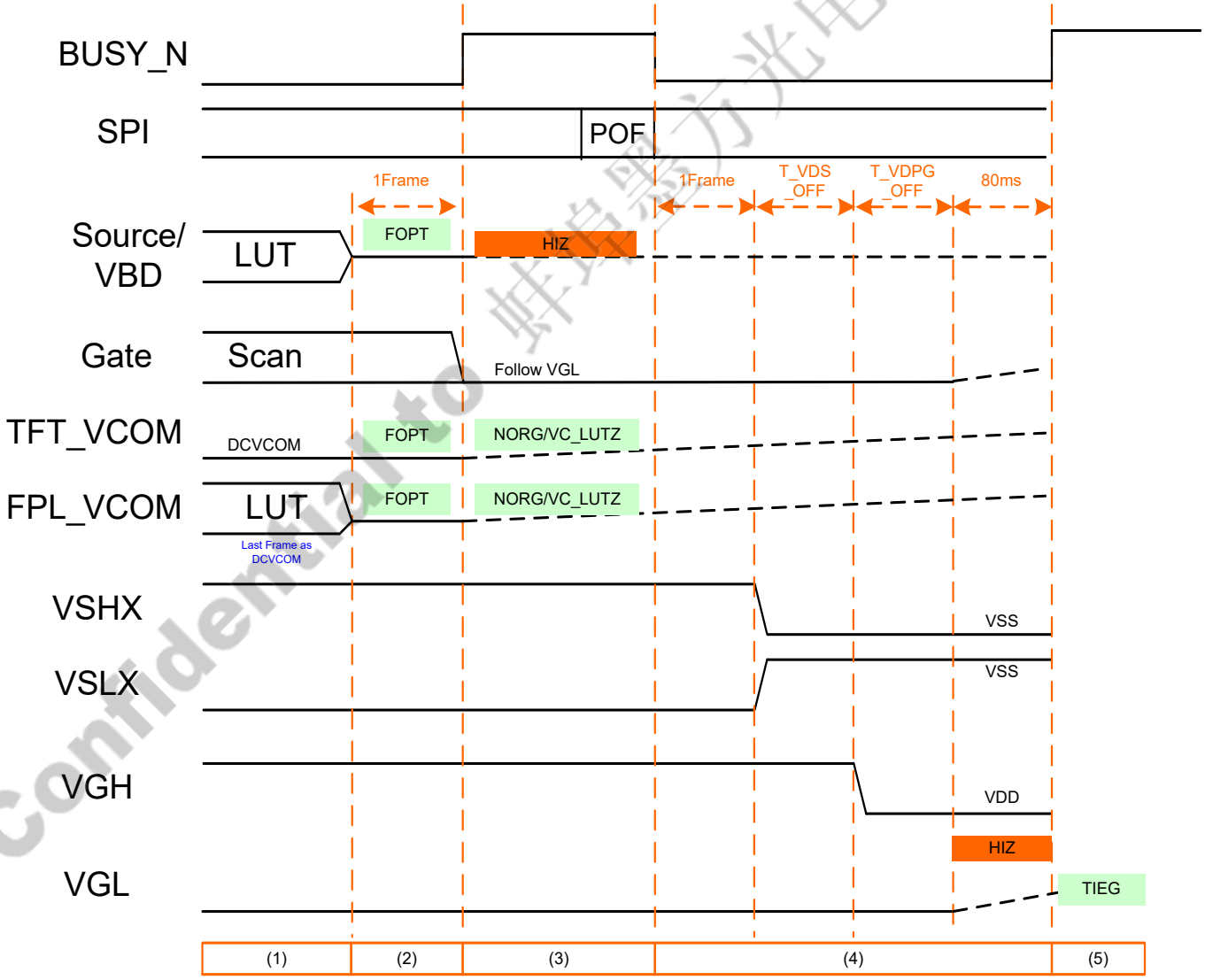


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9.2 Display on sequence from deep sleep mode



9.3 Display off Sequence



Note: If VCOM and VGL set as floating after display update. Please wait until the system completely discharge before next operation.

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10 Absolute Maximum Rating

Table 10-1 : Maximum Ratings

Symbol	Parameter	Rating	Unit
V_{DD}	Logic supply voltage	-0.5 to +4.0	V
V_{IN}	Logic Input voltage	-0.5 to V_{DDIO}+0.5	V
V_{OUT}	Logic Output voltage	-0.5 to V_{DDIO}+0.5	V
T_{OPR}	Operation temperature range	-30 to +85	°C
T_{STG}	Storage temperature range	-65 to +150	°C

Maximum ratings are those values beyond which damages to the device may occur. Functional operation should be restricted to the limits in the Electrical Characteristics tables or Pin Description section.

This device contains circuitry to protect the inputs against damage due to high static voltages or electric fields; however, it is advised that normal precautions be taken to avoid application of any voltage higher than maximum rated voltages to this high impedance circuit. For proper operation it is recommended that VDD be constrained to the range GND < VDD. Reliability of operation is enhanced if unused input is connected to an appropriate logic voltage level (e.g., either GND or V_{DDIO}). Unused outputs must be left open. This device may be light sensitive. Caution should be taken to avoid exposure of this device to any light source during normal operation. This device is not radiation protected.

11 Electrical Characteristics

The following specifications apply for: GND=0V, VDD=3.0V, T_{OPR}=25ºC.

Table 11-1: DC Characteristics

Symbol	Parameter	Applicable pin	Test Condition	Min.	Typ.	Max.	Unit
V_{DD}	VDD supply operation voltage	VDD		2.3	3.0	3.6	V
V_{DDD}	VDDD logic operation voltage	VDDD		1.7	1.8	1.9	V
V_{COM_DC}	VCOM_DC output voltage	TFT_VCOM		-4.0	-	-0.1	V
dV_{COM_DC}	VCOM_DC output voltage deviation	TFT_VCOM	No output load	-200	-	200	mV
V_{COM_AC}	VCOM_AC output voltage	FPL_VCOM		V_{SL} + V_{COM_DC}	V_{COM_DC}	V_{SH}+ V_{COM_DC}	V
V_{GATE}	Gate output voltage	G0~G599		-20	-	+20	V
$V_{GATE(p-p)}$	Gate output peak to peak voltage	G0~G599		-	-	40	V
VSH	Positive Source output voltage	VSH		-	+15	-	V
V_{SH_LV}	Positive Source output voltage	VSH_LV		+3	+11	+15	V
V_{SH_LV2}	Positive Source output voltage	VSH_LV2		+3	+5	+15	V
dV_{SH}	VSH/VSH_LV/VSH_LV2 output voltage deviation	VSH/ VSH_LV/VSL_LV2	No output load	-200	-	200	mV
V_{SL}	Negative Source output voltage	VSL		-	-15	-	V
V_{SL_LV}	Negative Source output voltage	VSL_LV		-15	-11	-3	V
V_{SL_LV2}	Negative Source output voltage	VSL_LV2		-15	-5	-3	V
dV_{SL}	VSL/VSL_LV/VSL_LV2 output voltage deviation	VSL/ VSL_LV/VSL_LV2	No output load	-200	-	200	mV
V_{IH}	High level input voltage	SI[0:3], SCL, CSB, D/C, RST_N, BS, M/S#, SYNC[0:1]		$0.8V_{DDIO}$	-	-	V
V_{IL}	Low level input voltage			-	-	$0.2V_{DDIO}$	V

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Symbol	Parameter	Applicable pin	Test Condition	Min.	Typ.	Max.	Unit
V_{OH}	High level output voltage	SI[0:3], BUSY, SYNC[0:1]	IOH = -100uA	$0.8V_{DDIO}$	-	-	V
V_{OL}	Low level output voltage		IOL = 100uA	-	-	$0.2V_{DDIO}$	V
V_{PP}	OTP Program voltage	VPP		7.25	7.5	7.75	V
Islp_VDD	Sleep mode current	VDD	- DC/DC off - No clock - No output load - MCU interface access - RAM data access	-	35	50	uA
Idslp_VDD	deep sleep current	VDD	- DC/DC off - No clock - No output load - No MCU interface access - Cannot retain RAM data	-	1	7	uA
Iopr_VDD	Operating Mode current	VDD	VDD=3V After PON Command VGH=20V VGL=-VGH VSH=15V VSH_LV=15V VSH_LV2=15V VSL_LV=-15V VSL_LV2=-15V VSL=-15V VCOM = -2V No waveform transitions. No output load No RAM read/write	-	6	-	mA
V_{GH}	Operating Mode Output Voltage	VGH		19.5	20	20.5	V
V_{SH}		VSH		14.9	15	15.1	V
V_{SH_LV}		VSH_LV		14.9	15	15.1	V
V_{SH_LV2}		VSL_LV2		14.9	15	15.1	V
V_{SL}		VSL		-15.1	-15	-14.9	V
V_{SL_LV}		VSL_LV		-15.1	-15	-14.9	V
V_{SL_LV2}		VSL_LV2		-15.1	-15	-14.9	V
V_{COM}		VCOM		-2.1	-2	-1.9	V

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12 AC Characteristics

12.1 Serial Peripheral Interface

The following specifications apply for: VDDIO - GND = 2.3V to 3.6V, T_{OPR} = 25°C, CL=20pF

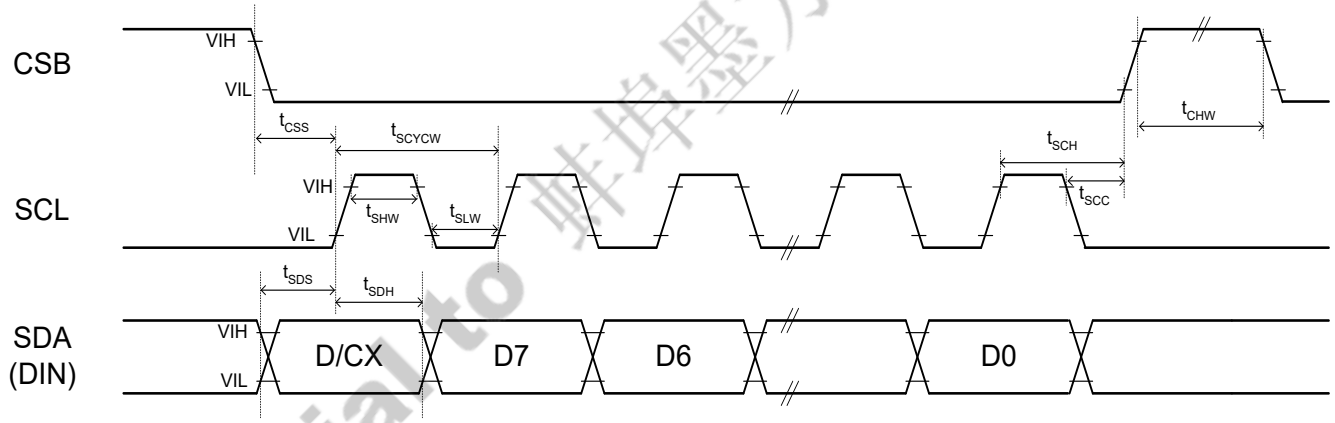
Table 12-1 : Serial Peripheral Interface Timing Characteristics

Write mode

Symbol	Parameter	Min	Typ	Max	Unit
t_{CSS}	CSB select setup time	60			ns
t_{CSH}	CSB select hold time	65			ns
t_{SCC}	CSB deselect setup time	20			ns
t_{CHW}	CSB deselect hold time	40			ns
t_{SCYCW}	Serial clock cycle (Write)	50			ns
t_{SHW}	SCL “H” pulse width (Write)	25			ns
t_{SLW}	SCL “L” pulse width (Write)	25			ns
t_{SCYCL}	Serial clock cycle (Read)	150			ns
t_{SHR}	SCL “H” pulse width (Read)	60			ns
t_{SLR}	SCL “L” pulse width (Read)	60			ns
t_{SDS}	Data setup time	30			ns
t_{SDH}	Data hold time	30			ns
t_{ACC}	Access time			75	ns
t_{OH}	Output disable time	10			ns

Note: All timings are based on 20% to 80% of VDDIO-GND

Figure 12-1: 3 pin serial interface characteristics (write mode)



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Figure 12-2: 3 pin serial interface characteristics (read mode)

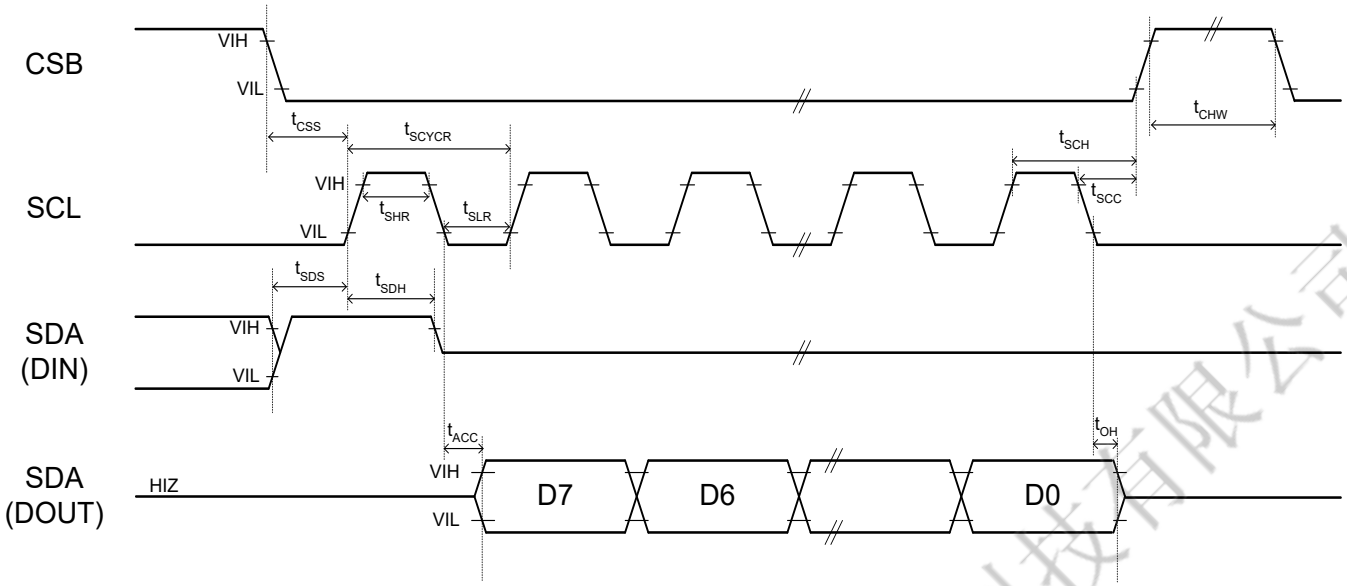


Figure 12-3: 4 pin serial interface characteristics (write mode)

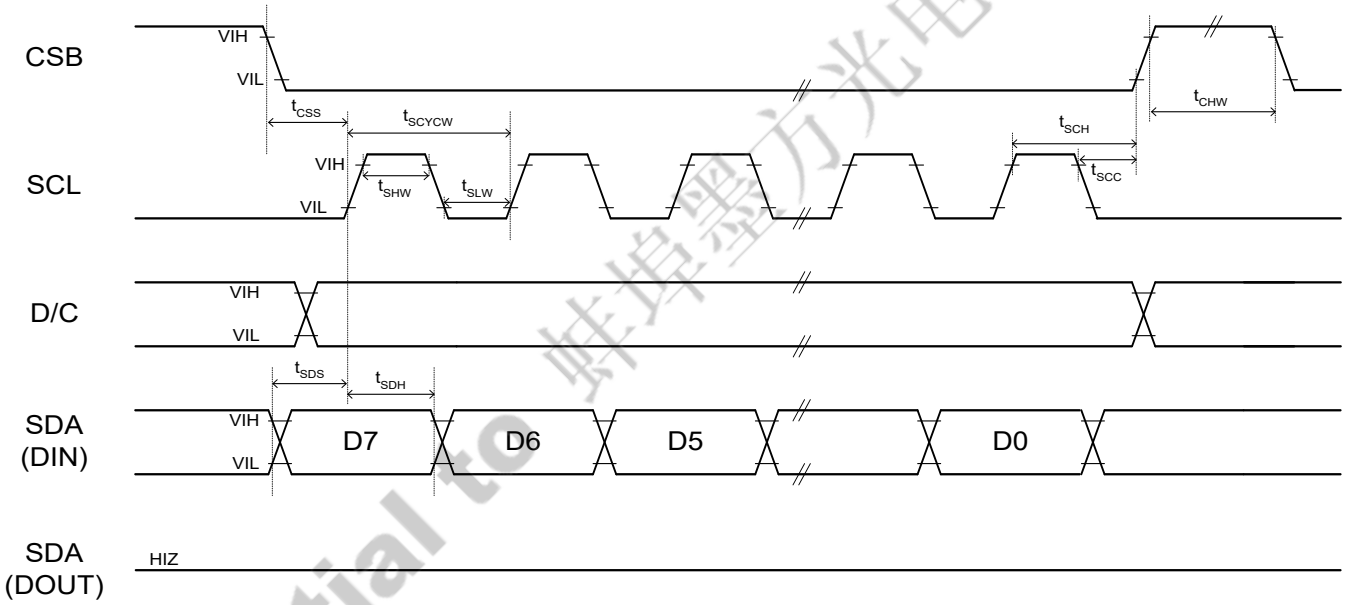
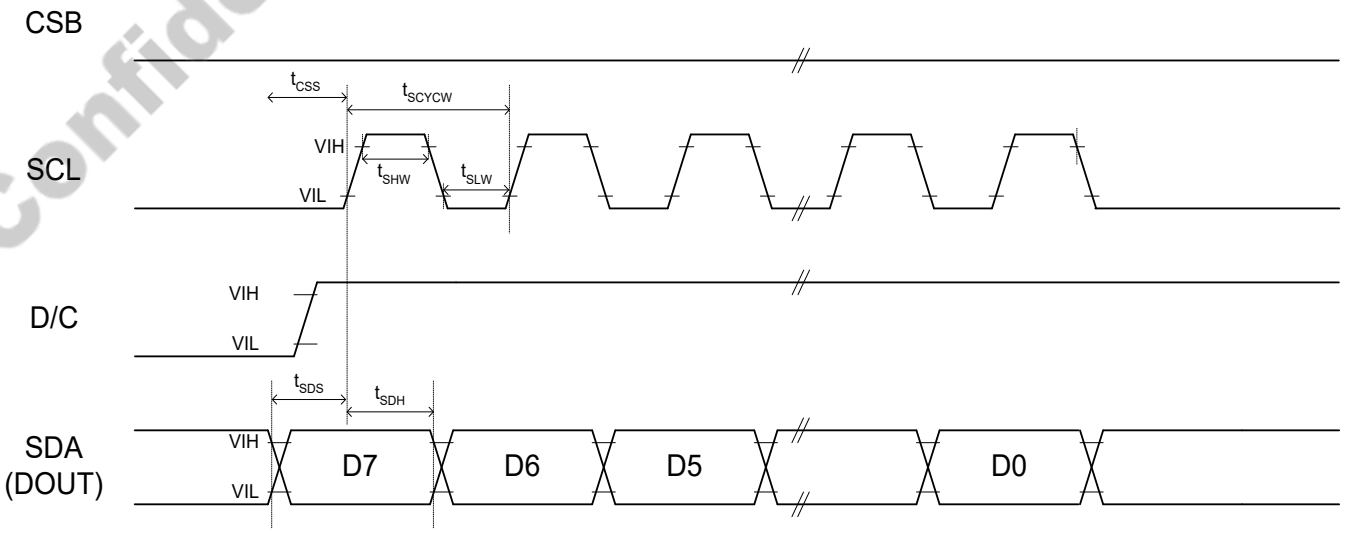


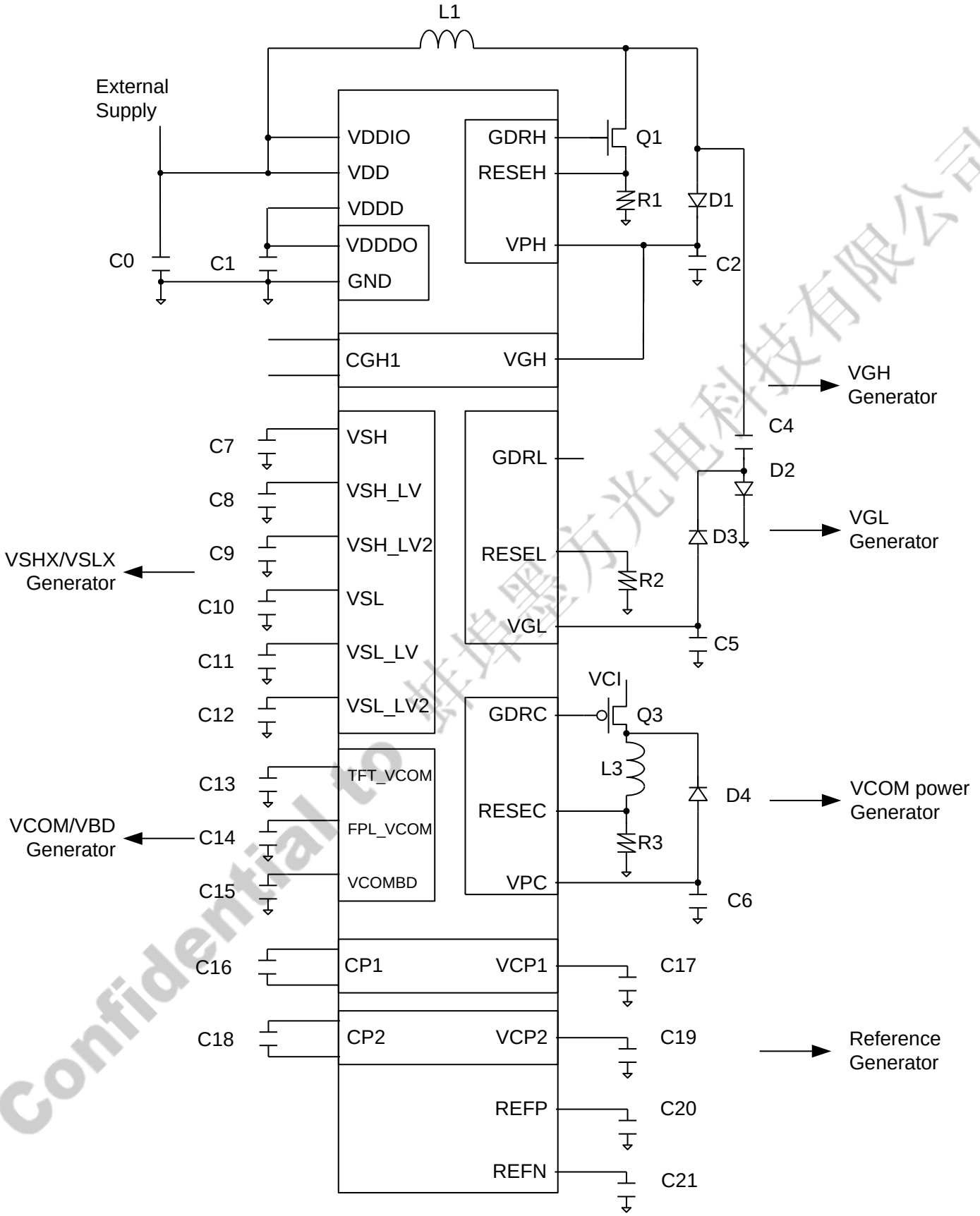
Figure 12-4: 4 pin serial interface characteristics (read mode)



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13 Application Circuit

Figure 13-1: SPD1657B application circuit



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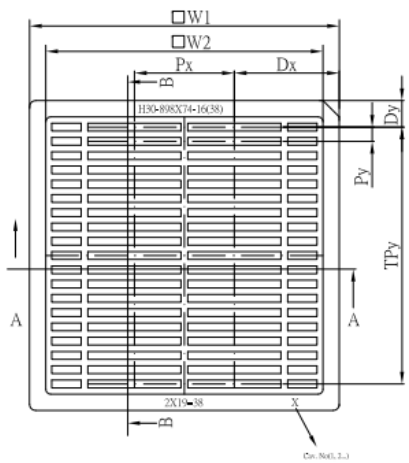
Table 13-1: Component list for SPD1657B application circuit

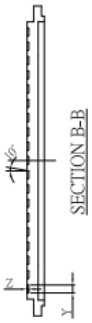
Part Name	Value	Requirement
C0	4.7uF	6V
C1	2.2uF	6V
L1	15uH	2A
Q1	NMOS	VDS=30V, Ron<=77mOhm @ 2.5V VGS,Typ Qg<= 3.5 nC
D1	Schottky Diode	30V
R1	0.2 ohm	0.2W
C2	10uF	25V, 1210, X7R
D2	Schottky Diode	30V
C4	10uF	25V, 1210, X7R
D3	Schottky Diode	30V
R2	0.15 / 0 Ohm	0.2W
C5	10uF	25V, 1210, X7R
L3	15uH	2A
Q3	PMOS	VDS = -20V, Ron<=85mOhm @ -2.5V VGS,Typ Qg<= 7.6nC
D4	Schottky Diode	20V
R3	0.22 ohm	0.2W
C6	10uF	25V, 1210, X7R
C7	10uF	25V, 1210, X7R
C8	10uF	25V, 1210, X7R
C9	10uF	25V, 1210, X7R
C10	10uF	25V, 1210, X7R
C11	10uF	25V, 1210, X7R
C12	10uF	25V, 1210, X7R
C13	4.7uF	10V, 0805, X5R/X7R
C14	0.47uF	25V, 1210, X7R
C15	0.47uF	25V, 1210, X7R
C16	1uF	10V, 0805, X5R/X7R
C17	1uF	10V, 0805, X5R/X7R
C18	1uF	10V, 0805, X5R/X7R
C19	1uF	10V, 0805, X5R/X7R
C20	1uF	10V, 0805, X5R/X7R
C21	1uF	10V, 0805, X5R/X7R

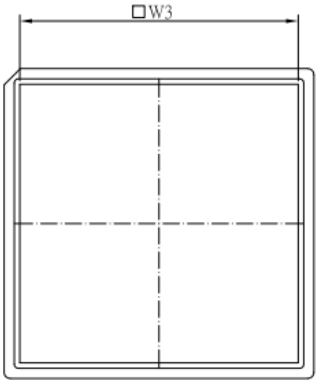
Remark: Component value is subjected to change and depends on panel loading.

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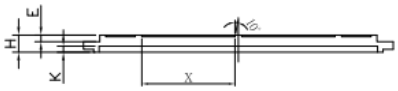
14 Package Information







Spec	
	mm (mil)
W1	76.00±0.10 (2992)
W2	68.00±0.10 (2677)
W3	68.30±0.10 (2689)
H	4.20±0.10 (165)
E	1.60±0.10 (63)
K	1.50±0.10 (59)
Px	24.50±0.10 (965)
Py	3.50±0.05 (138)
Dx	25.75±0.05 (1014)
Dy	6.50±0.05 (256)
TPy	63.00±0.10 (2480)
X	22.82±0.05 (898)
Y	1.88±0.05 (74)
Z	0.40±0.05 (16)
N	2x19=38(pocket number)



SECTION A-A

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